CHEMISTRY GRADE 10: THE PERIODIC TABLE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Inorganic Chemistry
Sub-Strand	Sub-Strand 1.3: The Periodic Table
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Development of the periodic table (Groups and periods) – Ion formation (oxidation number, valency, radicals) – Formulae and chemical equations – Chemical families: alkali metals, alkaline earth metals, halogens, noble gases, and transition elements
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) relate the position of an element in the periodic table to its electron arrangement, b) illustrate ion formation of elements, c) derive the formulae of compounds, d) write balanced equations for chemical reactions, e) appreciate the role of electron arrangement in the development of the periodic table.
Core Competencies	– Communication and Collaboration: The learner listens keenly and actively uses open questions while discussing with peers the relationship between atomic number, mass number, and number of electrons in atoms of elements — practised during group card-sorting activities and peer discussions on chemical families. – Critical Thinking and Problem Solving: The learner creates new ideas while drawing diagrams of ions of selected elements using dots and/or crosses to illustrate ion formation — applied when predicting cation and anion formation from electron arrangements. – Learning to Learn: The learner learns independently by writing balanced equations for chemical reactions — developed through guided practice and self-correction of chemical equations. – Citizenship: The learner appreciates contributions made by various scientists (e.g., Mendeleev, Newlands, Moseley) on the development of atomic structure and the periodic table.
Core Values	– Peace: The learner shows compassion as they display tolerance while predicting the type of ion formed from the electron arrangement of elements with peers — demonstrated during collaborative group sorting tasks and peer-review of ion diagrams.
Science & Engineering Practices	– Asking Questions and Defining Problems: Students generate questions about why elements behave differently (e.g., why sodium reacts violently with water but argon does not) and frame these as investigable problems tied to electron arrangement. – Developing and Using Models: Students build and iteratively revise particle-level models of ion formation and periodic trends,

	using dot-and-cross diagrams to represent electron loss and gain. – Analysing and Interpreting Data: Students analyse element card data (atomic number, mass, reactivity) to identify patterns across groups and periods, connecting data trends to electron configuration. – Obtaining, Evaluating, and Communicating Information: Students read and evaluate historical accounts of periodic table development (Döbereiner, Newlands, Mendeleev, Moseley) and communicate findings through group presentations and the Driving Question Board.
Pertinent & Contemporary Issues (PCIs)	– Socio-Economic and Environmental Issues (Social Cohesion): As the learner discusses with peers how atoms acquire stability, collaborative group norms and mutual respect are reinforced, mirroring the cooperative values needed in Kenyan community life. – Life Skills (Analytical Thinking): As the learner predicts the type of ion formed from a given electron arrangement of an atom, analytical and logical reasoning skills are developed that transfer to everyday decision-making.
Career Connections	– Pharmacist / Pharmaceutical Chemist: Understanding the periodic table, ion formation, and chemical formulae is foundational for drug formulation and interactions — relevant to Kenya's growing pharmaceutical sector (e.g., companies in Nairobi's Industrial Area). – Agricultural Chemist / Agronomist: Knowledge of element families (e.g., nitrogen, potassium, phosphorus) and their compounds underpins fertiliser use critical for Kenyan smallholder farmers in the Rift Valley and Central Kenya. – Materials Scientist / Metallurgist: Transition metals and their variable oxidation states are central to mining and materials processing, connecting to Kenya's titanium mining in Kwale County. – Environmental Chemist: Ion chemistry and chemical equations link to water quality testing relevant to managing Lake Victoria and Nairobi River pollution. – Secondary School Chemistry Teacher: Deep conceptual understanding of the periodic table is essential for educators who will shape the next generation of Kenyan scientists.
Focus for Lessons	This unit anchors learning in a puzzling real-world challenge: why do some everyday Kenyan substances (table salt, baking soda, bleach, charcoal) behave so differently even though they are all made of a small set of elements? Students investigate the historical and logical development of the periodic table, discover how electron arrangement determines an element's position and chemical behaviour, and use that understanding to explain ion formation, derive chemical formulae, and write balanced equations. The instructional approach is phenomenon-driven and student-centred — students sort, predict, model, and revise understanding across all 10 lessons, with every phase explicitly linked back to the anchoring phenomenon and the Driving Question Board.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do? KICD KEY INQUIRY QUESTION: 1. How do chemical bonds influence the properties of substances?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Mystery Element Card Challenge": Students are presented with a set of 20 unlabelled cards (one per element, H–Ca) each showing atomic number, mass number, melting point, reactivity rating, and a photograph of a Kenyan context where the element appears (e.g., sodium in table salt used to season ugali; chlorine in water treatment at Nairobi's Gigiri plant; calcium in bones of Maasai cattle; carbon in charcoal used for nyama choma). Students are asked to arrange the cards in any logical order that groups similar cards together. Initial attempts produce several different groupings across the class — some group by mass, some by reactivity, some by appearance. The puzzle: Why do certain elements — separated by many others in mass — behave almost identically, while elements sitting next to each other can be completely different? This mirrors exactly the confusion historical chemists faced before the periodic table was invented, and creates genuine need-to-know that drives the entire unit. The phenomenon is puzzling because students cannot yet

	explain the invisible internal structure (electron arrangement) that causes these visible patterns of behaviour.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Element Card Sort: Students encounter the 20 unlabelled element cards (H–Ca) showing Kenyan contexts. In groups they sort cards by any criteria they choose, share groupings, and articulate what is puzzling. The Driving Question Board (DQB) is started — students post "What we notice" and "What we wonder" sticky notes. Initial qualitative models are sketched: "What do you think makes these elements different on the inside?" Lesson 2 — History of the Periodic Table (Observe & Explain): Students watch a curated video on the historical development of the periodic table (Döbereiner's triads → Newlands' octaves → Mendeleev's predictions → Moseley's atomic number). Groups brainstorm on a timeline poster. Discussion focuses on: What problem were these scientists trying to solve? How is their challenge similar to our card-sort? DQB updated with new questions about atomic number vs. mass number ordering. Lesson 3 — Electron Arrangement and Periodic Table Position (Explain & Model): Students review electron configuration (s/p notation) for the first 20 elements. Teacher-guided activity: students write electron arrangements for H–Ca and physically place their element cards into a blank 4-period, 8-group grid drawn on the classroom floor (or large paper). Pattern discovered: elements in the same group have the same number of outer-shell electrons. Models revised to include electron shells on element card diagrams. Lesson 4 — Chemical Families (Observe & Explain): Students identify alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 17), noble gases (Group 18), and transition elements from their sorted grid. Using data cards showing reactivity and physical properties, students compare trends within groups and across periods. Kenyan contexts reinforced: potassium fertilisers in Rift Valley farms (Group 1); calcium in Maasai cattle bones (Group 2); chlorine in Mombasa water treatment (Group 17). DQB updated: "Why does reactivity increase down Group 1 but decrease down Group 17?" Lesson 5 — Ion Formation: Cations and Anions (Predict & Observe): Students predict (before instruction) whether sodium, magnesium, chlorine, and oxygen will lose or gain electrons to achieve stability, writing their predictions in notebooks. Teacher demonstrates dot-and-cross diagrams on the board for Na → Na^+ and Cl → Cl^-. Students then draw ion diagrams for K, Ca, O, and S using dots and/or crosses. Peer-check in pairs. Misconception addressed: atoms do not "want" electrons — stability is the driver (octet rule introduced contextually). Lesson 6 — Oxidation Numbers and Valency (Explain & Model): Students infer valency and oxidation numbers from the electron arrangements they wrote in Lesson 3. Teacher guides students to tabulate valencies for Groups 1, 2, 16, and 17 elements. Variable oxidation numbers introduced for transition elements (e.g., iron as Fe^{2+} in $FeCl_2$ and Fe^{3+} in $FeCl_3$ — linking to rust from Kisumu Port railings). Radicals (SO_4^{2-}, NO_3^-, CO_3^{2-}, OH^-, NH_4^+) introduced with their charges and valencies. DQB updated: "How do we know which oxidation number to use?" Lesson 7 — Deriving Chemical Formulae (Explain & Practice): Students use the valency/oxidation-number table from Lesson 6 to derive formulae of ionic compounds using the crossover method. Practice set: NaCl, MgO, $CaCl_2$, Al_2O_3, $FeSO_4$, $Fe_2(SO_4)_3$, $Ca(NO_3)_2$, NH_4Cl. All examples tied to Kenyan contexts: calcium nitrate fertiliser used in Kericho tea farms; ammonium chloride in dry-cell batteries common in Nairobi markets. Students self-mark against answer key and identify error patterns. Lesson 8 — Writing and Balancing Chemical Equations (Explain & Practice): Students are introduced to word equations, then symbol equations, then balanced equations using simple reactions: sodium + water → sodium hydroxide + hydrogen; magnesium + oxygen → magnesium oxide; calcium carbonate → calcium oxide + carbon dioxide (lime burning in Athi River). Step-by-step

	balancing strategy taught (atom inventory table). Students practise 6 equations independently, then peer-review. DQB updated: "Why must equations be balanced — what law does this connect to?" Lesson 9 — Consolidation & Extended Problem Solving (Apply & Model): Students receive a scenario card: "A Kenyan farmer in Nakuru wants to use three different fertilisers — urea (CH_4N_2O), calcium ammonium nitrate ($Ca(NO_3)_2 \cdot NH_4NO_3$), and muriate of potash (KCl). Identify the chemical families of each element present, write the formulae from valencies, and write a balanced equation for the reaction of potassium with water." Groups work collaboratively, present solutions, and critique each other's balanced equations. Models are revised to show the link between periodic table position → electron arrangement → ion formation → formula → equation. Lesson 10 — Model Gallery Walk, DQB Closure & Unit Reflection: Students produce a final annotated model poster that traces the storyline: element card (Lesson 1) → periodic table position → electron arrangement → ion formation → formula → balanced equation, using one Kenyan element/compound as a case study (e.g., sodium chloride from Malindi salt flats). Gallery walk with peer feedback using "Glow and Grow" sticky notes. DQB revisited: teacher facilitates whole-class discussion on which questions have been answered and which remain open (bridging to Sub-Strand 1.4 — Chemical Bonding). Unit assessed via a short written task: given an unknown element's electron arrangement, students predict its group, valency, ion type, and write the formula of its compound with chlorine.
Supporting Phenomena	– Sodium vs. Argon in Kenyan water treatment: Sodium (added as salt to food nationwide) reacts violently with water, yet argon (used in welding workshops in Nairobi's Industrial Area) is completely unreactive — why do two elements so close in mass behave so oppositely? (Supports: electron arrangement and noble gas stability) – Rusting of iron railings at Kisumu Port vs. gold ornaments at Maasai markets: Iron corrodes rapidly in the humid Lake Victoria environment while gold remains shiny for generations — both are transition metals, yet their reactivities are vastly different. (Supports: variable oxidation numbers and transition element behaviour) – Formation of salt crystals during sea-water evaporation at Malindi: Visible cubic crystals of sodium chloride form as Indian Ocean water evaporates — a striking macroscopic result of microscopic ion formation (Na^+ and Cl^-). (Supports: ion formation, valency, and chemical formulae) – Colour change of litmus in bleach vs. baking soda solutions used in Kenyan homes: Bleach (containing chlorine compounds) and baking soda (sodium hydrogen carbonate) produce opposite effects on indicators despite both being white powders — connecting halogen and alkali metal chemistry to observable household phenomena. (Supports: chemical families and balanced equations)

LESSON 1 (80 minutes): The Mystery Element Card Challenge — Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon by having learners sort 20 unlabelled element cards (H–Ca) and articulate what is puzzling about element behaviour, thereby generating genuine need-to-know that will drive the entire sub-strand.
Knowledge	Learners will know that the 20 elements from hydrogen to calcium can be organised by observable properties (atomic number, mass number, melting point, reactivity) and that some elements that are far apart in mass behave almost identically, while neighbouring elements can behave completely differently.
Skills	Learners will be able to: (i) sort and classify unlabelled element cards using self-chosen criteria; (ii) communicate and justify their grouping decisions to peers; (iii) record observations and questions on a Driving Question Board (DQB); (iv) sketch an initial qualitative model of what might cause differences between elements on the inside.
Attitudes	Learners appreciate that confusion and disagreement are normal starting points in science, showing tolerance and compassion (Peace value) while comparing different groupings with peers, and begin to appreciate the contributions of historical scientists to the development of the periodic table (Citizenship competency).
Key Inquiry Question	How do chemical bonds influence the properties of substances?
Purpose in Storyline	This is the opening lesson of the sub-strand. It plants the anchoring phenomenon — the Mystery Element Card Challenge — in learners' minds and opens the Driving Question Board. Every subsequent lesson will return to this card set as learners build the knowledge needed to fully explain why elements behave as they do. No answers are given today; the goal is productive confusion and rich questioning.
Safety Notes	No hazardous chemicals are used in this lesson. All materials are paper-based card sets and sticky notes. Ensure learners handle printed cards carefully and that the classroom floor is clear if groups move desks. If any card photograph shows a chemical context (e.g., chlorine water treatment), remind learners not to attempt to reproduce such processes at home.

B. LESSON OVERVIEW
Learners are presented with 20 unlabelled element cards (Hydrogen to Calcium) each displaying atomic number, mass number, melting point, reactivity rating, and a photograph of a Kenyan context where the element appears (e.g., sodium in table salt seasoning ugali; chlorine in water treatment at Nairobi's Gigiri plant; calcium in bones of Maasai cattle; carbon in charcoal for nyama choma; iron in a Kisumu metalworks; potassium in tea fertiliser from Kericho). Working in groups of four, learners sort the cards by any criteria they choose, then compare groupings across the class. The resulting disagreement — some groups sort by mass, some by reactivity, some by Kenyan use-context — surfaces the core puzzle: why do certain elements separated by many others in mass behave almost identically, while elements sitting next to each other can be completely different? The lesson closes with the opening of the Driving Question Board (DQB) and individual initial model sketches, setting up every lesson that follows in the sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Before seeing the element cards,	Distribute only the Carbon teaser	Individual written prediction	Teacher circulates during silent

📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	each learner individually examines a single 'teaser' card for Carbon (atomic number 6, mass number 12, melting point ~3 550 °C, reactivity rating: LOW, photograph: charcoal glowing under nyama choma on a Nairobi roadside grill). Learners write two individual predictions in their notebooks: (1) 'If I were given 19 more cards like this one — for different elements — what features would I use to group them together?' and (2) 'Do you think elements with similar masses always behave similarly? Why or why not?' Learners hold their predictions privately — they will revisit them after the card sort.	card (one per learner). Say: 'Look carefully at every piece of information on this card. You have 3 minutes — working completely alone and in silence — to write your two predictions in your notebook. There are no wrong answers; I want your honest thinking before we do anything else.' Set a visible 3-minute timer. After time is up, cold-call 3 learners (select one high-confidence, one mid, one quieter voice) using name cards or a random spinner. Ask each: 'Tell us ONE feature you would use to group elements, and WHY you chose that feature.' Apply 12-second WAIT TIME after each cold-call before accepting the answer. Record the three criteria on the board in a column headed 'Our Predicted Grouping Criteria' — do not evaluate them yet. Say: 'Keep these predictions visible. By the end of today you will know whether your instinct was on the right track — or whether the elements surprised you.'	(Predict-Before-Observe protocol). This activates prior knowledge from Grade 9 matter and particles, creates a cognitive anchor that will produce productive surprise during the card sort, and gives the teacher immediate formative data on what naïve frameworks learners bring to the periodic table.	writing and scans notebooks for the two predictions. Observable indicator: learner has written at least one specific, named property (e.g., 'mass', 'melting point', 'how reactive it is') rather than a vague statement. Learners who write only 'how it looks' are flagged for targeted questioning during the card sort phase.
Observe Phase 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each group of four receives a sealed envelope containing 20 unlabelled element cards (H–Ca). Cards are numbered 1–20 on the back only — no element names, no symbols. Each card face shows: Atomic Number, Mass Number, Melting Point (°C), Reactivity Rating (1 = very low to 5 = very high), State at room temperature, and a Kenyan context photograph with a one-line caption (examples: Card 3 — Lithium: 'Used in rechargeable batteries powering mobile phones sold at Nairobi's Gikomba market'; Card 11 — Sodium: 'Found in table salt used to season ugali at a	As groups open envelopes say: 'Your only rule is this — you must be able to EXPLAIN your grouping rule to someone else. Sort the cards in any way that makes sense to your group. You have 12 minutes. Go.' Circulate continuously, spending no more than 90 seconds at each group. Use these specific prompts when groups are stuck: 'What two cards feel most similar to you? Start there.' When a group sorts purely by mass/atomic number, probe: 'Cards 3 and 11 are far apart in number — look at their reactivity ratings. What do you notice?' Apply 12-second WAIT TIME.	Collaborative card sort with gallery walk (Evidence-Based Argumentation setup). The physical manipulation of cards externalises thinking and makes competing classification schemes visible to the whole class simultaneously. The gallery walk introduces the social dimension of scientific disagreement without confrontation — mirroring the historical debate between Mendeleev and Meyer, and connecting to the Citizenship core competency.	Teacher photographs each group's card arrangement on a phone/tablet before gallery walk begins. Observable indicators: (1) Every group has written a sticky-note grouping rule — not just arranged cards silently. (2) At least one group has spontaneously sorted by atomic number (indicating prior exposure). (3) At least one group has noticed the reactivity pattern without prompting (Mg, Ca both rated 3; Na, K both rated 5) — this group will be spotlighted in the Explain phase. Red flag: any group that has not moved cards at all after 4 minutes — teacher intervenes with the 'two

	home in Kisumu'; Card 17 — Chlorine: 'Used to purify drinking water at the Gigiri treatment plant, Nairobi'; Card 20 — Calcium: 'Strengthens the bones of Maasai cattle in the Rift Valley'). Groups have 12 minutes to sort the 20 cards into groups using ANY criteria they choose. They must: (a) physically arrange cards on the desk, (b) write their grouping rule on a sticky note placed above each group, and (c) note at least TWO observations that surprised them. After sorting, one representative from each group does a 90-second 'gallery walk' to observe one other group's arrangement without talking — returning to report back what was different.	When a group sorts by reactivity, probe: 'Does Card 2 (Helium, reactivity 1) really belong with Card 6 (Carbon, reactivity 1)? What else is different?' After gallery walks, cold-call 4 group reporters (one from each group): 'Tell us ONE thing another group did that was completely different from your group — and say whether it makes sense.' Record each unique grouping criterion on the board. Do NOT reveal which grouping is 'correct'. Say: 'Look at the board — we have [number] completely different ways to sort the same 20 cards. A German chemist named Lothar Meyer and a Russian chemist named Dmitri Mendeleev had exactly this argument in 1869. How do you think they resolved it?'		most similar cards' prompt.
Explain Phase 📹 **VIDEO:** [Modeling with systems of inequalities] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Writing the Periodic Table (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher reveals that each card represents one real element, and displays a projected image of the modern periodic table alongside a projected image of Mendeleev's original 1869 draft table. Learners are NOT yet told how the periodic table works — instead they are guided through a structured 'Notice and Compare' activity. Each group receives a 'Comparison Strip' — a printed list of three pairs of elements from their card set: Pair A (Li and Na), Pair B (F and Cl), Pair C (He and Ne). For each pair they must answer: 'These two elements are far apart in atomic number. List ALL the ways their cards are similar, and ALL the ways they are different.' Groups share findings. The teacher then introduces the vocabulary term PERIOD and GROUP using only the two projected images (no	Project both tables side by side. Say: 'You now know these are real elements. I am not going to tell you their names yet — you will earn those names over the next lessons. Right now I want you to think about these three pairs on your Comparison Strip.' Allow 6 minutes for group discussion of the three pairs. Cold-call 3 groups to share one similarity and one difference per pair — 9 sharings total, each with 10-second WAIT TIME before the group responds. After all three pairs are discussed, point to the projected modern periodic table and say: 'Scientists call these vertical columns GROUPS and these horizontal rows PERIODS. That is all I will tell you today.' Then ask the whole class (show of hands first, then cold-call 2 volunteers): 'Which column do you think Sodium and Lithium belong in — and what is	Structured Compare-and-Contrast with Constrained Vocabulary Introduction (Notice–Name–Hypothesise sequence). By withholding element names and symbols until learners have already grouped by property, the teacher ensures the classification logic is built from evidence rather than memorised from labels. The blank grid sketch forces spatial reasoning and previews the electron-arrangement explanations that will come in later lessons.	Collect the blank grid sketches at the end of the phase. Observable indicators: (1) Learner has placed Li and Na in the same column — shows transfer of the card-sort pattern to a 2D structure. (2) Learner's 'inside the element' sketch shows some particle-level thinking (circles, layers, dots) — indicates readiness for electron configuration lessons. (3) Learner's sketch is purely macroscopic (colour, texture drawings) — indicates the sub-atomic particle concept needs re-activation before Lesson 3. Teacher annotates sketches with a dot code (green dot = particle-level thinking present; orange dot = macroscopic only) for differentiation planning.

	definition given yet) and asks: 'Based on what you see — where do you think Li and Na sit relative to each other on this table?' Learners sketch their own rough placement of at least 5 cards onto a blank periodic table grid (provided as a half-page outline showing only the grid, no labels).	your evidence from the cards?' After responses, say: 'You have just done what Mendeleev did. You looked at behaviour patterns and used them to predict position. But here is the puzzle' — pause 12 seconds — 'Mendeleev did not know WHY elements in the same group behave similarly. Do you have a hypothesis? Sketch it — draw me what you think is INSIDE these elements that makes them act the same way. You have 4 minutes, working alone.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Modeling with systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives three sticky notes: one GREEN ('What I noticed'), one YELLOW ('What I wonder'), one PINK ('What I think explains it — my best guess right now'). Working individually and in silence for 4 minutes, learners write one statement per sticky note, as specifically as possible. Examples prompted on the board: GREEN — 'I noticed that Card 11 and Card 3 have very similar reactivity even though their atomic numbers are far apart'; YELLOW — 'I wonder why helium is unreactive but the element next to it in atomic number (lithium) is extremely reactive'; PINK — 'I think the NUMBER of something inside the atom — maybe a kind of layer — controls how reactive it is.' Learners then bring their sticky notes to the class DQB (a large sheet of chart paper or a dedicated section of the whiteboard divided into three columns: NOTICE / WONDER / BEST GUESS). The teacher reads 4 selected wonder-notes aloud and writes the most generative ones under the Driving Question in large text. The DQB remains visible on the wall for the entire unit.	Distribute sticky notes and say: 'Three minutes, silence, one idea per note — be as specific as you can. Use evidence from the cards. Write a card number if it helps.' After 3 minutes say: 'Stand up. Walk to the board. Place each note in its column. Read two notes that are NOT yours before you sit down.' Once all notes are placed, scan the WONDER column silently for 30 seconds, then read 4 aloud — selecting ones that point toward electron arrangement, periodicity, reactivity trends, and ion formation. Say: 'These four questions are going to be our compass for the next 30 lessons. Every time we answer one of them, we will come back here and mark it — but I predict new questions will keep appearing.' Write the unit Driving Question at the top of the DQB in bold: 'WHY DO THE 118 ELEMENTS BEHAVE SO DIFFERENTLY, AND HOW CAN KNOWING AN ELEMENT'S POSITION IN THE PERIODIC TABLE HELP US PREDICT WHAT IT WILL DO?' Ask the class: 'How many of your wonder-notes are actually asking a piece of THIS big question?'	Driving Question Board with three-colour sticky note protocol (Notice–Wonder–Hypothesise). This strategy makes the class's collective intellectual state visible and persistent. It gives quieter learners an equal voice (written contribution rather than verbal), models scientific habit of mind (distinguishing observation from inference from hypothesis), and creates accountability — the DQB will be revisited and updated at the start of every subsequent lesson.	Teacher photographs the completed DQB. Observable indicators: (1) NOTICE notes reference specific card properties (numbers, ratings) — not just 'some elements are reactive'. (2) WONDER notes are genuine questions with a question mark — not restatements. (3) BEST GUESS notes include at least one particle-level or structural idea — 'I think it has something to do with the number of electrons on the outside.' Red flag: all BEST GUESS notes are purely descriptive ('I think it depends on the mass') — teacher plans to open Lesson 2 with a targeted question about sub-atomic structure to bridge to electron arrangement.

		(Show of hands — expected: most hands up.) 'Good. That means you are already thinking like chemists.'		
Model Building Phase 📹 **VIDEO:** [Modeling with systems of inequalities] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Writing the Periodic Table (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their individual blank-grid sketches from Phase 3 and now add a 'Model Panel' — a small box next to their grid where they draw their initial mental model answering: 'What do you think is INSIDE two elements from the same group (e.g., the element on Card 3 and the element on Card 11) that makes them behave similarly?' They label their drawing with at least two annotations. They also write one sentence: 'My model predicts that if I found a NEW element with [property], it would go in [position] because...' These initial models are photographed by the teacher and stored as 'Model Version 1' — they will be revised at the end of Lessons 5, 10, 15, 20, and 30 as the sub-strand progresses. Learners write their name and today's date on the model sheet and place it in their science portfolio.	Say: 'You have seen the OUTSIDE of these elements — the properties on the cards. Now I want your best guess about the INSIDE. Draw it. Label it. It does not have to be correct — it has to be YOUR thinking right now, so that in future lessons you can see how your ideas have changed.' Allow 6 minutes for individual drawing and annotation. Circulate and use these specific prompts for learners who are stuck: 'You told me Card 11 and Card 3 behave similarly. Draw two circles — one for each. Now put something INSIDE each circle that explains why they act the same.' For learners who finish early: 'Add a third element from a DIFFERENT group to your model — how is its inside different from the first two?' After 6 minutes, cold-call 2 volunteers to hold up their model and describe it in one sentence (12-second WAIT TIME). Do NOT correct any model at this stage. Say: 'Every model on this table is a hypothesis. Hypotheses are not wrong — they are waiting for evidence. We will come back to every one of these models and revise them. Keep them safe.' Close the lesson by pointing to the DQB: 'Tomorrow we begin answering the FIRST of your wonder questions — we will look at how the periodic table was actually built, and you will decide whether Mendeleev's grouping matches yours.'	Initial Model Construction (Anchor Model protocol). Drawing an explicit initial model — even an incorrect one — is a core NGSS Science and Engineering Practice (Developing and Using Models). It forces learners to commit to a mechanistic explanation before instruction, creating the conceptual 'gap' that subsequent lessons will fill. Storing 'Model Version 1' and revisiting it at milestone lessons makes conceptual change visible and personally meaningful to each learner.	Teacher collects and photographs all Model Version 1 sheets before learners leave. Observable indicators: (1) Model includes at least one internal structural feature (rings, layers, dots, a nucleus) — not just an element symbol. (2) Model shows TWO elements side by side and identifies something SHARED — indicates learner is reasoning about periodicity at a particle level. (3) Learner's predictive sentence includes a specific property AND a specific position — e.g., 'If it reacts like sodium it would go in the same column.' Red flag: model shows only the element card data copied out (no internal structure attempt) — these learners will be seated next to particle-level thinkers in Lesson 2 for structured peer support.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your journal before the next session:

1. PHENOMENON UPTAKE — Did the card sort produce genuine puzzlement, or did learners quickly converge on atomic-number ordering? If convergence was fast, plan to use the reactivity-anomaly pairs (Li/Na vs. Be/Mg) more explicitly in Lesson 2 to deepen the puzzle.
2. DQB QUALITY — Count the WONDER sticky notes that reference internal/particle-level thinking versus purely macroscopic observations. A healthy DQB for Lesson 1 should have at least 30% particle-level wonders. If fewer, open Lesson 2 with a targeted provocation: 'We know WHAT elements do — but WHY do they do it? What could be inside?'
3. EQUITY OF VOICE — Did the same 4–5 learners dominate cold-call responses? Review your name-card rotation and ensure quieter learners are pre-selected for the first cold-call in Lesson 2.
4. MODEL VERSION 1 AUDIT — Sort Model Version 1 sheets into three piles: (A) particle-level internal structure, (B) macroscopic only, (C) blank or minimal. Pile B and C learners need a targeted sub-atomic particles recall activity at the start of Lesson 2 (5-minute retrieval task on protons, neutrons, electrons from Grade 9).
5. KENYAN CONTEXT RESONANCE — Note which card photographs generated the most discussion. Cards referencing everyday items (ugali salt, nyama choma charcoal, Kericho tea fertiliser) typically spark the most engagement — plan to use those specific elements as worked examples in the electron arrangement lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted 20 unlabelled element cards (H–Ca) — each showing a Kenyan context photograph — into self-chosen groups. Different groups produced different arrangements: some sorted by atomic number, some by reactivity rating, some by state at room temperature, and some by the Kenyan context shown on the card. Comparison of three element pairs (Li/Na, F/Cl, He/Ne) showed that elements far apart in atomic number can share very similar properties, while neighbouring elements can be completely different.
What did I learn?	The 20 elements from Hydrogen to Calcium can be arranged into a two-dimensional grid of GROUPS (vertical columns) and PERIODS (horizontal rows). Elements in the same group share similar properties even when their atomic numbers are far apart. Scientists disagreed about how to organise the elements — just as our class disagreed — until patterns in behaviour made certain groupings more defensible than others. The unit Driving Question is: 'Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?'
How does this explain the phenomenon?	We cannot yet explain WHY elements in the same group behave similarly — that explanation requires understanding what is INSIDE atoms (electron arrangement), which we will build over the next lessons. Our initial models (Model Version 1) are our best current hypotheses. They are not wrong — they are waiting for evidence.

LESSON 2 (80 minutes): History of the Periodic Table: Standing on the Shoulders of Giants	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To trace the historical development of the periodic table and connect each scientist's contribution to the card-sort puzzle students experienced in Lesson 1.
Knowledge	Students will know the key contributions of Döbereiner, Newlands, Mendeleev, and Moseley to the development of the periodic table, and understand why atomic number — not mass number — became the organising principle.
Skills	Students will construct a annotated group timeline poster, sequence historical events chronologically, and update the Driving Question Board with evidence-informed questions.
Attitudes	Students will appreciate the contributions made by various scientists — across different nations and eras — to the development of the periodic table, recognising that scientific knowledge is built collaboratively over time (Citizenship value).
Key Inquiry Question	How do chemical bonds influence the properties of substances?
Purpose in Storyline	In Lesson 1, students physically grappled with the same puzzle that historical chemists faced: how do you impose order on 20 seemingly unrelated element cards? Lesson 2 reveals that real scientists — Döbereiner, Newlands, Mendeleev, and Moseley — each proposed a different ordering logic, just as different student groups did. This lesson validates students' card-sort strategies as genuine scientific thinking and reveals why atomic number ultimately unlocked the pattern, creating urgent need-to-know about what atomic number actually represents internally (setting up Lesson 3 on electron arrangement).
Safety Notes	No hazardous materials are used in this lesson. If printed cards or scissors are used for the timeline activity, remind students to handle scissors safely. Ensure the classroom projector/screen is positioned so all learners can view the video without eye strain.

B. LESSON OVERVIEW

Learners watch a curated, narrated video tracing the historical development of the periodic table from Döbereiner's triads (1817) through Newlands' Law of Octaves (1864), Mendeleev's predictive table (1869), and Moseley's atomic number revolution (1913). In groups, students construct annotated timeline posters that map each scientist's strategy onto the card-sort strategies their own class used in Lesson 1. A whole-class gallery walk and structured discussion surfaces the key insight: ordering by atomic number — not mass — reveals a deeper, invisible pattern. The lesson closes with a DQB update, where new questions about what atomic number physically means are posted, directly launching Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Periodic Table of Elements] Source: Khan	Students individually recall their Lesson 1 card-sort grouping strategy. They write a silent written prediction in their science notebooks (3 minutes): 'Before	1. Display the original class card-sort groupings on the board (photographs taken in Lesson 1) and say: 'Look at our four different groupings from last lesson. Keep	Silent Written Prediction followed by Turn-and-Talk. This activates prior knowledge from Lesson 1, lowers the affective filter before video viewing, and creates	Observable indicator: Every student has written at least one sentence in their notebook within the 3-minute window. Teacher scans the room during writing.

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	atomic number was discovered, which property do you think scientists used to organise the elements — and why would that cause problems?' Students then share with a shoulder partner for 2 minutes before a brief whole-class share-out. A volunteer from each group reads out their Lesson 1 grouping strategy (mass, reactivity, or appearance) to anchor the historical parallel. Students are explicitly told: 'Today you will find out that real scientists made almost exactly the same choices you did — and hit the same walls.'	these in mind — you are about to meet the scientists who had the same argument 200 years ago.' 2. Prompt the silent written prediction: 'You have 3 minutes — no talking, pens moving. What property did early scientists use, and what problem would that cause?' 3. After 3 minutes, give WAIT TIME of 10 seconds after saying 'Pens down,' then cold-call 3 students from different groups using the class register. Ask each: 'What property did you predict, and what's one reason it might fail?' 4. Record the three predicted properties on the board in three columns. Do NOT confirm or correct yet — say: 'Hold those predictions. The video will either support or challenge them.'	cognitive dissonance that the video will then resolve. By anchoring predictions to the class's own card-sort strategies, students approach the history not as passive recipients but as co-investigators.	During cold-call, listen for whether students reference atomic mass (correct historical parallel) or reactivity/appearance (alternative but valid reasoning). Flag any student who predicts 'atomic number' — probe them: 'What is atomic number, exactly?' — to surface prior knowledge gaps that will be addressed in the video.
Observe Phase 📹 **VIDEO:** Periodic Table of Elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students watch a 12-minute curated video on the history of the periodic table. The video covers: (i) Döbereiner's Triads — groups of three elements with similar properties, middle element's mass ≈ average of the other two (e.g., Li, Na, K); (ii) Newlands' Law of Octaves — every eighth element repeats in properties, compared to musical octaves, ridiculed by the Chemical Society; (iii) Mendeleev's 1869 table — ordered by atomic mass, with deliberate gaps for undiscovered elements, and his famous prediction of eka-silicon (Germanium); (iv) Moseley's 1913 X-ray work — reordering by atomic number resolves all anomalies (Co/Ni, Ar/K). Kenyan context is embedded: the video narrator explicitly links sodium (Na) — found in the salt shakers on every nyama choma table in Nairobi — to Döbereiner's Li-Na-K triad, and chlorine (Cl) — used to treat water	1. Before pressing play, distribute the structured observation guide and say: 'As you watch, your job is a detective's job — fill in this table. Every time a scientist appears, pause your thinking and ask: Is this the same problem our class had on the card sort?' 2. Pause the video at the 5-minute mark (after Newlands) and give 60 seconds for students to compare their observation guides with their neighbour. Say: 'Do not change your answers — just notice if you missed anything.' 3. Pause again after Mendeleev (approximately 9 minutes) and cold-call 2 students: 'What gap did Mendeleev leave, and why was that a brave scientific decision?' Allow WAIT TIME of 12 seconds before accepting an answer. 4. After the video ends, give 2 minutes for students to complete any empty cells in their guide, then say: 'Circle the one scientist whose ordering strategy	Structured Observation Guide (a form of Claim-Evidence-Reasoning scaffold). By requiring students to record the problem with each method — not just the method itself — the guide forces analytical viewing rather than passive absorption. Pausing the video at key moments creates micro-sensemaking checkpoints that prevent cognitive overload and connect video content back to the anchoring phenomenon in real time.	Observable indicator: Completed observation guide with all four columns filled for at least three of the four scientists by the end of the video. Teacher circulates during the two pause moments and looks specifically at the 'One problem with their method' column — this column reveals depth of understanding. A common misconception to watch for: students writing that Mendeleev's table was 'wrong' because it had gaps. If heard, say: 'Was leaving a gap a mistake — or a prediction? What's the difference between a mistake and a prediction in science?'

	at Nairobi's Gigiri plant — to the halogen family pattern. While watching, students complete a structured observation guide (4 columns): Scientist | Time period | How they ordered elements | One problem with their method.	is most similar to what YOUR group did in the card sort. Be ready to explain why.' 5. Cold-call 4 students (one per lesson-1 grouping strategy) to share their circled scientist.		
Explain Phase 📹 **VIDEO:** [Periodic Table of Elements] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Writing the Periodic Table (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four construct an annotated timeline poster (A2 size) with the following required elements: (a) a horizontal timeline from 1800–1920 with each scientist pinned at the correct date; (b) a speech bubble for each scientist summarising their ordering idea in one sentence, written as if the scientist is speaking (e.g., 'I, Mendeleev, arranged the 63 known elements by atomic mass and left gaps for elements not yet discovered — I was so confident I even predicted their properties!'); (c) an arrow from each scientist's entry pointing to the relevant card(s) from Lesson 1 that their method would have correctly grouped; (d) a red cross on any card their method would have placed in the wrong position or left unexplained. Groups then do a 5-minute gallery walk to view two other groups' posters before returning to their seats for a whole-class discussion. Discussion anchor question: 'All four scientists were intelligent and careful. Why did it take nearly 100 years to get from Döbereiner to Moseley — and what finally broke the deadlock?'	1. Assign group roles before the activity: Timeline Architect (draws the line and pins dates), Speech-Bubble Writer (writes the scientist's voice), Card Connector (draws arrows to Lesson 1 cards), and Devil's Advocate (finds the red-cross errors). Say: 'Every role matters — the Devil's Advocate is the most important scientist in the room right now.' 2. Circulate during poster construction. At each group, ask one probing question: 'Why did Newlands' octave pattern break down after calcium?' or 'Which two elements in our card set would Mendeleev have placed in the wrong order, and why?' Allow WAIT TIME of 10 seconds per probe. 3. During the gallery walk, give students one sticky note each and instruct: 'Leave one question or one challenge on a poster that is not your own.' 4. Return to whole-class discussion. Cold-call 3 students using the register: 'What sticky note question did your group receive that you couldn't answer?' Record these unanswered questions on the board — they will migrate to the DQB. 5. Deliver the key explanation: 'Moseley's breakthrough was not just a better number — it was a window into the atom's interior. Atomic number = number of protons. That means it connects directly to the invisible internal structure we haven't explained yet. That is what Lesson 3 is about.'	Scientist Voice Role-Play within Poster Construction. Writing in the first person as a historical scientist forces students to internalise the scientist's reasoning rather than merely report it. The Devil's Advocate role institutionalises critical thinking and directly mirrors the scientific practice of peer review (NGSS Science and Engineering Practice 7: Engaging in Argument from Evidence). The gallery walk with sticky-note challenges generates authentic peer-to-peer critique and surfaces genuine questions for the DQB.	Observable indicator: Each poster has at least one red-cross card correctly identified with a written explanation. During circulation, listen for students who can articulate WHY atomic number resolves anomalies (e.g., Co and Ni) rather than simply stating that it does. If a group is stuck, use the prompt: 'Look at cobalt and nickel on your cards — what does the mass number tell you? What does the atomic number tell you? Which one predicts their behaviour better?' Flag groups whose red-cross annotations are blank or incorrect for targeted follow-up during the DQB phase.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Periodic Table of Elements] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Writing the Periodic Table (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives three sticky notes (different colours: green = something we now understand, orange = a new question this lesson raised, pink = a connection to the original card-sort challenge). Groups have 4 minutes to write and post their notes on the class DQB (a large wall display established in Lesson 1). The teacher then reads out all orange (new question) sticky notes aloud and the class votes by show of hands for the top three most burning questions. These three are circled on the DQB and assigned a lesson number (e.g., 'Lesson 3 will answer this'). Likely emergent questions: 'What IS inside the atom that gives it an atomic number?', 'Why do elements in the same column of the table behave the same — what do they share?', 'Did any African or Kenyan scientists contribute to the periodic table?' The teacher validates the last question and notes it as a research extension task.	1. Distribute sticky notes and say: 'Three notes, three colours, three minutes. Green: one thing you now understand about why our card-sort had different answers. Orange: one new question this lesson opened up. Pink: one specific connection between what a historical scientist did and what your group did in Lesson 1.' 2. After posting, read every orange note aloud with energy: 'This is real science — every one of these questions was asked by a real chemist at some point.' 3. Run the vote for top three questions. Say: 'Hands up — which question, if answered, would most help you understand the card-sort puzzle?' Count hands publicly. 4. Map the top three questions to upcoming lesson numbers on the DQB: 'Question 1 will be answered in Lesson 3. Question 2 — Lesson 4. Question 3 — let's hold that one and see if the evidence leads us there.' 5. Close the phase by pointing to the card-sort display and asking: 'Has any group changed their mind about how they would sort the cards now? Thumbs up if yes.' If hands go up, ask one student to explain their revised thinking in one sentence.	Three-Colour Sticky Note DQB Protocol (a structured variant of the Know-Wonder-Learn framework). Using three distinct colours forces metacognitive differentiation — students must actively separate what they know from what they wonder and what they can connect. The public vote on questions creates shared intellectual ownership of the inquiry and makes the DQB a living document rather than a static display. This directly embeds NGSS Science and Engineering Practice 1: Asking Questions.	Observable indicator: Every group has posted at least one sticky note in each colour category. Teacher reads and photographs all sticky notes immediately after class. The orange (new question) notes are the highest-value formative data — they reveal conceptual gaps and naive theories (e.g., 'Is the atomic number just the mass divided by 2?') that must be addressed in Lesson 3. Pink notes reveal whether students have successfully connected the historical narrative back to the anchoring phenomenon — the key conceptual goal of this lesson.
Model Building Phase 📹 **VIDEO:** [Periodic Table of Elements] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Each student returns to their individual Lesson 1 element card model sketch (a rough diagram they drew in Lesson 1 showing how they would arrange a subset of 10 cards). They now make Revision 1 of their model: using a different coloured pen, they re-sort their 10 cards by atomic number and annotate the diagram with two sentences: (a) 'I changed [X] because Moseley's evidence showed that…' and (b) 'My model	1. Return Lesson 1 model sketches (or direct students to their notebook page). Say: 'This is your scientific model. Today you have new evidence. Scientists revise their models when they get new evidence — that is not weakness, that is science. Use a different colour pen so we can see your thinking evolve.' 2. Read the two sentence starters aloud from the board and say: 'Sentence one celebrates what you now	Annotated Model Revision with Colour-Coded Evidence Tracking. Using a second pen colour to show revisions makes the evolution of scientific thinking visible and tangible — a core practice in science (NGSS Science and Engineering Practice 2: Developing and Using Models). The deliberately incomplete second sentence activates forward-looking curiosity and ensures the model remains a 'work	Observable indicator: Revised model sketch shows cards re-ordered by atomic number (not mass) with at least one annotated change justified by reference to Moseley. The incomplete Sentence 2 annotation is the richest formative data point — it reveals whether students have identified the right next question (electron arrangement / internal structure). A student whose Sentence 2 reads 'I don't know'

[Writing the Periodic Table (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	still cannot explain why elements in the same column behave similarly, because…' The incomplete second sentence is intentional — it plants the seed for electron arrangement (Lesson 3). Students who finish early are invited to look up whether any East African scientists have contributed to chemistry research and to note one name and contribution.	understand. Sentence two is deliberately left open — it is your next investigation question. Do not try to answer it yet.' 3. Give WAIT TIME of 12 seconds after reading the prompts before students begin writing, to allow thinking. 4. Circulate and read Sentence 2 annotations. Cold-call 2 students to share their incomplete sentence: 'What do you think the answer MIGHT be — even if you're not sure?' After each response, say: 'Hold that thought — Lesson 3 will test it.' 5. Collect the revised model sketches (or photograph them) for teacher review. These serve as the primary formative artefact for Lesson 2.	in progress' rather than a closed product, which is epistemologically accurate — scientists' models are always provisional. This directly sets up the electron arrangement investigation in Lesson 3.	has not yet made the conceptual leap; a student whose Sentence 2 reads 'I still can't explain why Na and K react the same way even though their masses are very different — what do they share inside?' is ready for Lesson 3. Teacher uses these annotations to group students for Lesson 3's differentiated investigation.

D. TEACHER REFLECTION

After the lesson, review the three formative artefacts in order of priority: (1) Orange DQB sticky notes — do students' new questions point toward electron arrangement and internal atomic structure? If most questions are still about mass or surface properties, Lesson 3 will need a stronger bridge. (2) Revised model sketches — is Sentence 2 specific enough to drive inquiry, or is it vague? Vague Sentence 2 responses suggest students watched the video but did not connect it to the phenomenon. Plan a 5-minute re-anchor at the start of Lesson 3. (3) Observation guide 'One problem' column — were students able to critique each scientist's method, or did they only describe it? If critique is absent, the gallery walk sticky-note challenge protocol needs to be more explicitly modelled in Lesson 3. Additionally, note whether the Kenyan context (sodium in nyama choma salt, chlorine in Gigiri water treatment) landed as relevant — if students showed engagement spikes at those moments (leaning in, commenting), amplify this strategy in Lesson 3 by adding a Kenyan context hook at the start. If the extension question about East African scientists in chemistry was raised, prepare a brief slide for Lesson 3 opener featuring Dr. Julius Kipkemoi (Kenyan chemist, University of Nairobi) or other relevant regional scientists to honour the Citizenship competency.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched a curated video on the historical development of the periodic table (Döbereiner → Newlands → Mendeleev → Moseley) and completed a structured observation guide tracking each scientist's ordering method and its limitations. Groups constructed annotated timeline posters connecting each historical strategy to their own Lesson 1 card-sort approaches, and conducted a gallery walk with peer-challenge sticky notes.
What did I learn?	Students learned that the periodic table was not invented in a single moment but evolved over nearly 100 years as scientists proposed, tested, and revised different ordering systems. They learned that Mendeleev's use of atomic mass allowed brave predictive gaps, but that Moseley's use of atomic number — the number of protons — finally resolved all anomalies and revealed a deeper, invisible pattern inside the atom. They connected this historical arc to their own card-sort strategies from Lesson 1.
How does this explain the	Students cannot yet fully explain WHY ordering by atomic number works — specifically, what internal feature of the atom (electron

phenomenon?	arrangement) causes elements in the same group to behave similarly. This gap is intentionally preserved in their Sentence 2 model annotation and on the DQB, and will be directly investigated in Lesson 3.

LESSON 3 (80 minutes): Electron Arrangement and Periodic Table Position	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To help learners connect the internal electron arrangement of the first 20 elements (H–Ca) to their physical position in the periodic table, revealing that group number = number of outer-shell electrons — the invisible rule that explains why their Mystery Element Cards cluster the way they do.
Knowledge	Learners will know that: (a) electrons occupy energy levels (shells) described using s and p notation; (b) the period number indicates the number of occupied electron shells; (c) the group number (for Groups I–VIII) indicates the number of outer-shell (valence) electrons; (d) elements in the same group share the same valence electron count, which explains their similar chemical behaviour.
Skills	Learners will be able to: (a) write the electron arrangement of any of the first 20 elements (H–Ca) using s and p notation; (b) use electron arrangement to determine the correct period and group position of an element; (c) physically place element cards into the correct cell of a blank 4-period, 8-group grid; (d) annotate element card diagrams with shell models showing electron distribution.
Attitudes	Learners appreciate that an invisible internal structure — electron arrangement — is the hidden logic behind the visible patterns on the periodic table, and show tolerance and compassion while collaborating with peers to place cards correctly, reflecting the Peace value embedded in the curriculum.
Key Inquiry Question	How do chemical bonds influence the properties of substances?
Purpose in Storyline	In Lessons 1–2, learners struggled to sort their Mystery Element Cards by mass or reactivity alone and discovered that Mendeleev arranged elements by atomic number and repeating properties. Lesson 3 is the 'Explain & Model' pivot: learners now gain the microscopic explanation — electron shells — that finally makes sense of why sodium (Na) and potassium (K) behave so similarly despite being far apart in mass, and why chlorine (used in Nairobi's Gigiri water-treatment plant) is so different from argon sitting right next to it. By the end of this lesson, every element card has a shell diagram on its back, and the class floor-grid IS the periodic table — built by the learners themselves.
Safety Notes	No hazardous chemicals are used in this lesson. All activities are pencil-and-paper or card-based. Learners moving around the classroom floor grid should be reminded to walk carefully to avoid tripping on large paper sheets. Ensure the floor area is clear of bags and obstacles before the floor activity begins.

B. LESSON OVERVIEW

Lesson 3 is the central 'Explain & Model' lesson in the evidence-gathering sequence. After two lessons of observing patterns from the outside (card groupings, historical development, atomic number trends), learners now drill into WHY those patterns exist. Using s/p notation, learners write electron configurations for all 20 elements (H–Ca), then physically walk their element cards to the correct cell in a large blank grid drawn on the classroom floor or on 2 m × 1.5 m paper. The moment of discovery comes when learners notice — without being told — that every card they place in Group I has exactly 1 outer-shell electron, every Group VII card has 7, and the Noble Gas cards all have full outer shells. Teacher-guided Socratic questioning then links this back to the anchoring phenomenon: the mystery of why Na and K behave alike is solved. The lesson closes with learners revising their element card diagrams to include shell models, physically updating the artefact that anchors the whole unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two pairs of Mystery Element Cards face-down (only atomic number visible): Card Na (Z=11) and Card K (Z=19); Card Ne (Z=10) and Card Ar (Z=18). Teacher asks: 'Before we look at any other data — predict whether each pair will behave similarly or differently, and write ONE reason for your prediction in your notebook.' Learners write individual predictions (2 min silent think). Pairs then share with a shoulder partner (1 min). Four volunteers share whole-class. Predictions are written on the board under two columns: 'Similar behaviour' and 'Different behaviour' — these stay visible all lesson as a public record to be confirmed or revised.	1. Display the four cards on the board with only atomic number visible. Say: 'Do not look anything else up. Use only what you already know. In your notebook, predict: will Na and K behave similarly or differently? Will Ne and Ar behave similarly or differently? Write your reason.' SET A 12-SECOND WAIT. 2. Cold-call FOUR students (two who predicted 'similar', two who predicted 'different') — explicitly say: 'I am choosing people who wrote different reasons, so we can hear multiple ideas.' 3. Record all four predictions verbatim on the board. Say: 'We will return to these predictions at the end of the lesson. Do not erase them.' 4. Bridge statement: 'The only way to confirm or refute these predictions is to look INSIDE the atom. That is what we do today.'	Strategy — Predict-Then-Anchor: Articulating a prior prediction before instruction creates cognitive dissonance when evidence contradicts it. Recording predictions publicly on the board makes the intellectual stakes visible and motivates learners to pay attention to the Observe and Explain phases that follow. This strategy also surfaces misconceptions (e.g., 'K is heavier so it must be more different from Na') that the teacher can explicitly address.	Observable indicator: Every learner has a written prediction with a reason in their notebook before any pairs share. Teacher circulates during the 2-minute silent write and scans notebooks — look specifically for learners who leave the 'reason' blank (prompt: 'What is ONE thing you remember about Na or K from Lesson 1?'). Note the proportion of 'similar' vs 'different' predictions for each pair — this reveals the class's current mental model entering the lesson.
Observe Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Writing Electron Configurations (s/p notation): Learners receive a structured worksheet listing all 20 elements (H–Ca) with atomic number provided. Working in pairs, they write the full electron configuration in s/p notation (e.g., Na: $1s^2\ 2s^2\ 2p^6\ 3s^1$). A laminated reference card showing the Aufbau filling order (1s → 2s → 2p → 3s → 3p → 4s) is on each desk. Teacher models the first three (H, He, Li) live on the board, narrating aloud. Pairs complete the remaining 17 elements (15 min). PART B — Identifying Outer-Shell Electrons: Using a highlighter,	PART A: 1. Model H ($1s^1$), He ($1s^2$), Li ($1s^2\ 2s^1$) on the board, saying: 'I am filling the lowest energy level first. Watch how I read this: $1s^2$ means 2 electrons in the first shell s-subshell.' 2. Release pairs to work. Circulate; after 3 minutes cold-call ONE pair: 'Read me your configuration for Carbon.' Verify aloud. 3. After 8 min, pause whole class: 'Hold your pens. Does anyone have a question about Neon or Argon specifically?' Take 2 hands. PART B: 4. After highlighter step, say: 'Look	Strategy — Embodied Cognition through Physical Mapping: Physically walking to a grid on the floor and placing a card engages kinesthetic and spatial reasoning alongside symbolic notation. Research shows that embodied experiences (moving through space) deepen retention of abstract concepts. The floor grid becomes a shared, manipulable model — not a diagram in a textbook — reinforcing that the periodic table is a human-constructed tool for organising evidence, which directly mirrors the anchoring phenomenon's 'card-sorting' challenge. The gallery walk layer adds a second	Observable indicators: (1) During Part A, teacher checks that configurations end correctly — common error: writing $3p^6$ for Ar then forgetting $4s^2$ for Ca. Listen for pairs self-correcting using the reference card. (2) During Part B, the written 'outer-shell electron' column is the key artefact — collect 5 random worksheets after Part B to scan for accuracy before Part C. (3) During floor placement, each pair must state period and group with a reason before placing — a student who says 'I just counted' without referencing shells is a red flag; redirect immediately. (4) Gallery walk notebook entries are reviewed at the start of Phase

	learners circle the last term in each configuration (the outermost subshell electrons). They record in a second column: 'Number of outer-shell electrons.' Teacher pauses the class after 5 elements are highlighted and asks: 'What do you notice about the outer-shell electron count for Li and Na? Don't tell your partner yet — just write it.' PART C — Floor Grid Placement: The teacher (with student helpers, set up during the predict phase) has taped a 4-row × 8-column grid on the classroom floor using masking tape, labelled 'Period 1–4' on rows and 'Group I–VIII' on columns. Each pair receives their element cards from Lessons 1–2. Pairs take turns walking to the grid and placing their card in the cell they believe is correct based on period (number of shells) and group (outer-shell electrons). The class observes each placement; anyone who disagrees raises a hand silently. After all 20 cards are placed, the class does a gallery walk around the grid.	only at Li and Na. Write in your notebook what you notice about their outer-shell electrons. 10-second wait.' Cold-call TWO students. 5. Then ask: 'Now compare Na and K. Same question. 10-second wait.' Cold-call TWO different students. PART C (Floor Grid): 6. Demonstrate placement of H (Period 1, Group I) and He (Period 1, Group VIII/0) yourself, narrating: 'H has 1 shell — Period 1. It has 1 outer electron — Group I. I place it here.' 7. Each pair approaches the grid, states their element name, period, group, and reason BEFORE placing. If incorrect, teacher says: 'Interesting — tell me the number of shells first.' Do NOT give the answer; redirect to their worksheet. 8. After all cards placed, say: 'Walk silently around the grid for 2 minutes. In your notebook write: one pattern you see in any column, and one question the grid raises for you.'	pass of observation, allowing learners to notice group-column patterns they may have missed while focused on their own element.	3 by cold-calling 3 students to share their 'one pattern'.
Explain Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES	PART A — Pattern Articulation (Think-Pair-Share): Learners look at the completed floor grid and answer in writing: 'What do all elements in Group I have in common in their electron arrangement? Group VII? Group VIII?' They share with their pair, then four pairs share whole-class. Teacher scribes answers on the board. The class co-constructs the rule: 'Group number = number of outer-shell (valence) electrons; Period number = number of	PART A: 1. Point to Group I column on the floor grid. Ask: 'Without touching the cards, call out the outer-shell electron count for every card in this column.' Record the numbers on the board. 12-second wait between each element. Ask: 'What is the pattern?' Cold-call TWO students. 2. Repeat for Group VII and Group VIII/0. 3. Say: 'In your notebook, write the rule in ONE sentence — no more.'	Strategy — Rule Co-Construction followed by Application Transfer: Rather than stating the group/period rule directly, the teacher elicits it from learners by pointing to the floor grid evidence they built themselves (Phase 2). This inductive approach — pattern → rule — mirrors scientific reasoning (NGSS Practice 6: Constructing Explanations). The Kenya-context application tasks (Maasai cattle bones, Gigiri plant, welding Argon) immediately	Observable indicators: (1) The 3-sentence explanation in Part B is the primary written formative artefact — teacher collects 6 random notebooks and reads during Part C group discussion; look for: correct use of 'outer-shell electrons', connection to group number, and a causal link to behaviour (not just correlation). (2) In Part C, monitor group discussions — a group that discusses only the element name without referencing electron

for similar readings	occupied electron shells.' PART B — Connecting to the Phenomenon: Teacher reveals the full data on the Na and K cards and the Ne and Ar cards (from the Predict phase). Learners revisit their predictions. Guided discussion: 'Na has 1 outer electron. K has 1 outer electron. They are in the same group. How does this explain why they react similarly with water?' Learners write a 3-sentence explanation in their notebooks connecting electron arrangement → group position → similar behaviour. PART C — Kenya Context Application: Teacher presents three new scenarios: (1) Calcium (Ca) in the bones of Maasai cattle — 'Why is Ca in Group II? Count its outer electrons.' (2) Chlorine (Cl) used at Nairobi's Gigiri water-treatment plant — 'Why is Cl in Group VII? Why is it so reactive?' (3) Argon (Ar) — 'Why is Ar used in light bulbs and welding instead of nitrogen?' Learners discuss in groups of 3 for 4 minutes, then each group shares one answer. PART D — Core Competency: Critical Thinking — Learners are posed a challenge: 'A new element is discovered. Its electron configuration is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$. Without looking at any table, tell me: what period, what group, and predict whether it will be a metal, metalloid, or non-metal.' Learners write individual responses (3 min), then cold-call 3 students.	10-second write. Cold-call THREE students; select the most precise wording and say: 'That is the scientific rule. Everyone write that exact sentence and underline it.' PART B: 4. Return to the prediction board. Say: 'Look at your prediction from Phase 1. Was it correct? Why or why not? Write 3 sentences connecting electron arrangement to your answer.' 12-second wait. Cold-call TWO students who had OPPOSITE predictions — ask both to explain using electron arrangement. 5. Explicitly name the conceptual link: 'This is why Mendeleev's table works — not magic, not mass, but valence electrons. The invisible explains the visible.' PART C: 6. For each Kenya scenario, say: 'Group discussion — 4 minutes. I will cold-call ONE person from each group, chosen at random. Make sure everyone in your group can answer.' After 4 min, cold-call 3 groups. 7. For Argon, explicitly connect: 'Full outer shell = no need to gain or lose electrons = unreactive = safe for use near hot metals in welding.' PART D: 8. Read the configuration aloud. Say: 'Individual work only. 3 minutes. Write your period, group, and reasoning.' Cold-call 3 students. If any student gives the wrong group, ask: 'Count the electrons in the last subshell only	transfer the rule to authentic situations learners recognise, preventing the rule from remaining abstract. The novel-element challenge in Part D is a transfer task that assesses whether learners own the rule or are merely pattern-matching.	arrangement has not yet internalised the rule; redirect with: 'Tell me the electron configuration first.' (3) Part D individual response: correct answers are Period 3, Group V, non-metal — any student answering Period 2 has likely miscounted shells; note for differentiated follow-up.

		— what do you get?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes ONE sticky note to add to the class Driving Question Board. The sticky note must be in one of three colours: GREEN = 'Something I can now explain about the Mystery Element Cards that I could not explain in Lesson 1'; ORANGE = 'A new question this lesson raised for me'; RED = 'Something I am still confused about.' Learners post their notes in the correct section of the DQB. Teacher reads all red and orange notes aloud (anonymously) and the class votes (thumbs up) for which questions to investigate in Lessons 4–5. Two questions are circled as 'Priority Questions for Next Lesson.' The DQB is photographed and displayed digitally on the ARES platform.	1. Distribute sticky notes (3 colours). Say: 'You have 3 minutes. Write exactly ONE sticky note. It must be a complete sentence — not just a word.' Set a visible 3-minute timer. 2. As learners post notes, circulate and read them. Identify 2 green notes that best capture today's key learning and read them aloud: 'I want to share two green notes — listen for the scientific language.' 3. Read ALL red notes aloud (anonymously). For each, ask the class: 'Can anyone in the room partially answer this now?' Take 1 hand per red note. 4. Say: 'I am circling two orange questions that connect most directly to our driving question. We will attack these in Lesson 4.' Circle them visibly with a marker. 5. Photograph the DQB. Say: 'This board is your evidence that science is built by questions, not just answers — exactly like Mendeleev's notebook.'	Strategy — Three-Colour Metacognitive DQB: Differentiating sticky notes by colour forces learners to metacognitively categorise their own understanding state (confident, curious, confused) rather than producing a generic 'what I learned' statement. The public posting and class voting on orange questions gives learners genuine agency over the lesson sequence, reinforcing the KICD 'Learning to Learn' core competency. Connecting the DQB to Mendeleev's notebook of questions sustains the historical storyline thread and reminds learners that confusion is a productive scientific state.	Observable indicators: (1) Every learner posts exactly one sticky note — any learner who has not posted after 4 minutes is approached individually: 'Tell me one thing you can explain now that you couldn't in Lesson 1.' Scribe it for them if needed. (2) The ratio of green:orange:red gives the teacher a whole-class understanding profile. If more than 30% of notes are red, the opening of Lesson 4 must include a 10-minute re-teach of the group/period rule before advancing. (3) Quality check: green notes should reference 'outer-shell electrons' or 'valence electrons' — vague green notes ('I understand more') signal surface learning only.
Model Building Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar	Learners retrieve their physical Mystery Element Cards from the floor grid (each pair takes back their own cards). On the BACK of each card, they draw a Bohr shell diagram showing the electron arrangement for that element — concentric circles with electron counts per shell. Below the diagram they write the s/p configuration and label: 'Outer-shell electrons: ___'. Cards for Group I elements (H, Li, Na, K) are lined up side by side on the desk; learners annotate: 'All have 1 outer electron → same group → similar behaviour.' The pair then writes a	1. Say: 'Turn your element card over. You are now going to upgrade your card — you are adding the hidden layer that explains everything we saw in the grid. Draw the Bohr model. Use the rule: shell 1 holds max 2 electrons, shell 2 holds max 8, shell 3 holds max 8 for our 20 elements.' 2. Model Na on the board: draw 3 concentric circles, write 2 in shell 1, 8 in shell 2, 1 in shell 3. Say: 'This ONE electron in the outer shell — that is the whole story of sodium's reactivity.' 3. Circulate. For any incorrect	Strategy — Cumulative Artefact Revision (Model Upgrading): The physical act of flipping the element card and adding a new layer (Bohr diagram + s/p notation + Element Story) makes the model revision tangible and cumulative. Learners can literally see both sides of the card — observable properties (front) and microscopic explanation (back) — embodying the science practice of connecting macroscopic phenomena to atomic-level models (NGSS Practice 2: Developing and Using Models). The 'Element Story' writing task integrates literacy,	Observable indicators: (1) Bohr diagram accuracy — shell counts must be correct for all 20 elements; the most common error is K (2,8,8,1 — not 2,8,9). Teacher specifically checks all K and Ca cards before they are returned to envelopes. (2) Element Story must contain all four required elements (Kenya context, period, group, outer-shell count, behaviour prediction) — award a quick star on the card back for complete stories during circulation. (3) Exit check: as learners pack away cards, teacher holds up a random element symbol (e.g., Mg) and

readings	revised 'Element Story' for their element (2–3 sentences): e.g., 'Sodium (Na) is used to season the ugali eaten across Kenya. It sits in Period 3, Group I, because it has 3 shells and 1 outer electron. This single outer electron makes it highly reactive — it easily gives it away to form Na^+.' Cards are returned to their envelope for use in Lessons 4–10.	diagram, ask: 'How many total electrons does this element have? Start filling from the innermost shell.' Do NOT correct directly — redirect to the filling rule. 4. After diagrams are complete (8 min), say: 'Now write the Element Story. It must include: the element's Kenyan context from the card, its period and group, its outer-shell electron count, and ONE prediction about its behaviour.' 5. Cold-call THREE pairs to read their Element Story aloud. After each, ask the class: 'Does this story use evidence from today? Thumbs up if yes.' 6. Close by saying: 'Your card now has two layers — the outside (what we can observe) and the inside (electron arrangement). That is the same two-layer thinking every chemist uses. In Lesson 4, we use this inside layer to explain ion formation — how elements lose or gain those outer electrons to become charged.'	requires causal reasoning ('because … therefore …'), and ensures the Kenya context is not cosmetic but embedded in the scientific explanation.	asks the whole class to write on a scrap of paper: period, group, outer-shell electrons. Collect scraps — this is the lesson exit ticket. Correct answer: Period 3, Group II, 2 outer electrons.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) FLOOR GRID LOGISTICS — Did the grid placement run smoothly, or did crowding slow the activity? If more than 5 minutes were lost to logistics, consider assigning elements to pairs before the lesson rather than self-selecting. (2) CONCEPTUAL BREAKTHROUGH MOMENT — Did learners spontaneously notice the group-column pattern, or did you have to name it for them? If you named it first, plan a more structured discovery prompt for Lesson 4's analogous activity (e.g., 'Write down the outer-shell electron count for every card in ONE column before looking at the group label'). (3) RED STICKY NOTES — What were the most common confusions? If many learners are confused about the difference between period number and shell number, open Lesson 4 with a 5-minute targeted re-teach using only Na (2,8,1) and K (2,8,8,1) as worked examples. (4) ELEMENT STORIES — Did learners use causal language ('because', 'therefore') or merely descriptive language ('sodium has 1 outer electron')? If predominantly descriptive, introduce a sentence frame in Lesson 4: 'Because [element] has ___ outer-shell electrons, it [behaviour], which is why it is used in [Kenyan context].' (5) KICD COMPETENCY EVIDENCE — Note which learners demonstrated Communication and Collaboration (active listening during floor placement discussions) and Critical Thinking (the novel-element Part D challenge) — record initials for portfolio evidence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students placed 20 element cards (H–Ca) onto a large floor grid (4 periods × 8 groups) after writing s/p electron configurations,

	and conducted a gallery walk to observe patterns across rows and columns.
What did I learn?	The group number equals the number of outer-shell (valence) electrons, and the period number equals the number of occupied electron shells — the invisible internal structure that explains why Na and K behave alike, and why Cl (Gigiri water treatment) and Ar (welding) behave so differently despite being neighbours.
How does this explain the phenomenon?	Elements in the same group share the same number of outer-shell electrons, which drives their similar chemical behaviour — the hidden rule that resolves the Mystery Element Card sorting puzzle introduced in Lesson 1.

LESSON 4 (80 minutes): Chemical Families: Identifying Groups and Trends in the Periodic Table	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To identify the major chemical families (Groups 1, 2, 17, 18, and transition elements) in the periodic table and compare trends in reactivity and physical properties within groups and across periods.
Knowledge	Students will know the names and positions of alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 17), noble gases (Group 18), and transition elements, and describe trends in reactivity and physical properties within these families.
Skills	Students will identify chemical families from the sorted element grid, interpret data cards to compare properties, and construct evidence-based explanations for trends within groups and across periods.
Attitudes	Students appreciate the role of electron arrangement in causing observable patterns of behaviour across chemical families, and show tolerance and collaboration while discussing trends with peers (Peace value).
Key Inquiry Question	How do chemical bonds influence the properties of substances?
Purpose in Storyline	In Lessons 1–3, students physically sorted the Mystery Element Cards and began mapping elements into a grid by atomic number and period. In Lesson 4, they now zoom in on vertical columns — chemical families — and use data cards to explain why potassium fertilisers power Rift Valley crops, why chlorine purifies Mombasa's water, and why noble gases are used in safety lighting. This lesson provides the first evidence that position in the periodic table predicts behaviour, directly advancing the driving question about why elements behave so differently.
Safety Notes	No hazardous chemicals are used in this lesson. All activities use printed data cards and reference materials. If teacher demonstrates a Group 1 metal reaction video, ensure volume is appropriate and students understand they must never handle alkali metals directly. Remind students that chlorine gas (Group 17) used in water treatment is toxic in concentrated form — discuss only in the context of safe, dilute water-treatment applications.

B. LESSON OVERVIEW
Students work with their sorted Mystery Element Card grids from Lessons 1–3 and a new set of Chemical Family Data Cards to identify and characterise five major chemical families: alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 17), noble gases (Group 18), and transition elements. Through structured group analysis, whole-class data sharing, and a cross-group comparison chart, students construct evidence-based explanations for why reactivity increases down Group 1 (linking to Kenyan potassium fertilisers in the Rift Valley) but decreases down Group 17 (linking to chlorine in Mombasa water treatment). Students update the Driving Question Board with new, more precise questions about electron arrangement as the root cause of these trends. The lesson closes with a cumulative model revision in which students annotate their element grid with colour-coded family labels and directional trend arrows.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students examine their sorted Mystery Element Card grids from	Distribute the Chemical Family Prediction Slips face-down. Say:	Prediction Activation (Anticipatory Sensemaking): By requiring written	Observable indicator: Every student has at least three of four

📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Lessons 1–3. Each student receives a 'Chemical Family Prediction Slip' with four sentence frames: (1) 'I think the elements in column ___ are a family because…'; (2) 'I predict that as you go down Group 1, reactivity will ___ because…'; (3) 'I predict that as you go down Group 17, reactivity will ___ because…'; (4) 'I think noble gases are special because…'. Students write individual predictions silently for 5 minutes, drawing on the Kenyan context clues on the element cards (e.g., potassium card shows Rift Valley fertiliser bags; chlorine card shows the Gigiri water treatment plant in Nairobi; calcium card shows Maasai cattle bones). Predictions are not collected — students keep them to compare against evidence gathered later in the lesson.	'Before we look at any new data, I want your brain's best guess. Flip your slip over — you have 5 minutes, working completely on your own. Use the clues on your element cards.' Circulate silently for 3 minutes, noting 3–4 contrasting predictions to use later. At the 4-minute mark, cold-call 3 students: 'Achieng, what did you write for Group 1 reactivity trend?' — allow 10 seconds WAIT TIME before prompting. After 3 responses, say: 'Interesting — we have at least two different predictions about Group 17. Let's gather evidence and find out who is right, and more importantly, WHY.' Do not correct any prediction at this stage.	predictions before evidence is introduced, students activate prior knowledge from Lessons 1–3 and create a cognitive commitment that makes subsequent evidence more memorable. The Kenyan context images on the cards give students anchoring cues that tie abstract chemical trends to familiar real-world materials (fertiliser bags, water treatment, cattle bones), lowering the entry barrier for all learners.	sentence frames partially completed before the group activity begins. Teacher scans slips during circulation — look specifically for whether students attempt a 'because…' justification or leave it blank. Blank 'because' sections flag students who need additional scaffolding in the Observe phase. Note at least 2 contrasting Group 17 predictions to use as productive disagreement in Phase 3.
Observe Phase 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 receive a set of 12 Chemical Family Data Cards (pre-printed, laminated). Each card shows: element name, symbol, atomic number, group number, period number, melting point (°C), reactivity rating (1–5 scale), state at room temperature, one distinctive property, and a Kenyan context image. The 12 cards cover: Li, Na, K (Group 1); Mg, Ca (Group 2); F, Cl, Br (Group 17); He, Ne, Ar (Group 18); Fe, Cu (transition elements). Task 1 (10 min): Groups physically sort the 12 cards into family columns on their desk, matching them to the corresponding columns in their existing element grid. Task 2 (12 min): Groups complete a structured 'Trend Detective Table' with four columns — Family Name, Property That Changes Going	Distribute data card sets and Trend Detective Tables simultaneously. Say: 'Task 1: sort these 12 cards into family columns — you have 10 minutes. Task 2: fill in your Trend Detective Table — you have 12 minutes. Task 3: one surprising question per group on a sticky note.' Set a visible countdown timer. At the 8-minute mark of Task 1, if groups are struggling, prompt: 'Look at the group number printed on each card — does that help you decide which column it belongs in?' During Task 2, circulate and ask targeted questions to 2–3 groups: 'You wrote reactivity increases down Group 1 — which Kenyan context card on your table is the MOST reactive alkali metal? What does that tell us?' Allow 10 seconds WAIT TIME. For groups	Data Card Sorting with Trend Detection: Physical manipulation of laminated cards makes the abstract concept of 'chemical families' tangible and spatial. The Trend Detective Table scaffolds students toward the NGSS Science and Engineering Practice of Analysing and Interpreting Data by requiring them to identify both the direction of a trend and a real-world Kenyan example that makes it meaningful. The sticky-note question task prepares content for the DQB phase and gives the teacher real-time insight into what students find genuinely puzzling.	Observable indicator: Each group's Trend Detective Table has at least 3 rows completed with a direction arrow and a Kenyan example before Phase 3 begins. Teacher checks specifically: (a) Does the Group 1 row show reactivity INCREASING downward? (b) Does the Group 17 row show reactivity DECREASING downward? (c) Is the Kenyan example column populated with a specific context (not just 'fertiliser' but 'potassium fertiliser used by maize farmers in the Rift Valley')? Groups with incorrect trend directions are flagged for targeted questioning in Phase 3.

	Down the Group, Direction of Change (↑/↓), and One Kenyan Example of This Property in Use. Groups must populate at least 3 rows. Task 3 (5 min): Each group selects their single most surprising finding and writes it as a question on a sticky note for the Driving Question Board.	that finish early: 'Can you identify a trend in melting points going down Group 1? Add it to your table.' Cold-call one student per group during a 3-minute whole-class share-out: 'Kamau, what direction did your group find for Group 17 reactivity going down?' (10 seconds WAIT TIME). Record responses on the board in a class-wide summary table.		
Explain Phase 📹 **VIDEO:** [Naming ions and ionic compounds](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Writing the Periodic Table (PLIX)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Whole class reconvenes. The teacher reveals a large printed or projected 'Chemical Families Summary Chart' showing the periodic table grid with Groups 1, 2, 17, 18, and transition elements colour-coded. Students use their Trend Detective Tables as reference. Activity 1 — Claim-Evidence-Reasoning (CER) Writing (12 min): Each student individually writes a CER response to the prompt: 'Explain why potassium (K) is more reactive than lithium (Li), even though both are in Group 1.' Sentence starters are provided: Claim — 'Potassium is more reactive than lithium because…'; Evidence — 'According to our data cards, Li has a reactivity rating of ___ and K has ___…'; Reasoning — 'This pattern makes sense because as you go down Group 1…'. Activity 2 — Cross-family Contrast Discussion (8 min): Teacher facilitates a structured whole-class discussion comparing Group 1 and Group 17 reactivity trends. Students refer back to their Prediction Slips from Phase 1. Discussion anchor: 'Chlorine is used to purify water at the Mombasa treatment plant — but fluorine, above it in Group 17, is even more reactive. Bromine,	Display the CER prompt on the board. Say: 'This is individual writing time — 12 minutes, no talking. Use your Trend Detective Table and your data cards as evidence.' After 12 minutes, cold-call 3 students to read their CER aloud — choose one strong example, one partial example, and one that has a claim but missing reasoning. For each, ask the class: 'What evidence did they use? Is the reasoning connected to the trend we observed?' Allow 10 seconds WAIT TIME after each question before taking responses. For the cross-family discussion, pose: 'Raise your hand if your Phase 1 prediction for Group 17 was CORRECT.' Then: 'Raise your hand if you were surprised.' Say: 'Njoroge, you predicted Group 17 reactivity would increase going down — what evidence changed your mind?' (10 seconds WAIT TIME). Explicitly connect to the driving question: 'We can now partially answer our driving question — position in the table tells us something about reactivity. But WHY does position predict reactivity? What is happening INSIDE the atom? That is what we still need to explain.' Write this on the board as a bridge to upcoming lessons on electron arrangement.	Claim-Evidence-Reasoning (CER) Writing: CER structures student thinking by separating the conclusion (claim) from the supporting data (evidence) and the logical bridge between them (reasoning). This directly mirrors the NGSS Science and Engineering Practice of Constructing Explanations. The cross-family contrast discussion uses productive disagreement — students whose Phase 1 predictions were wrong must publicly reconcile prediction with evidence, which deepens encoding. Connecting back to the anchoring phenomenon (the Mystery Element Cards) reminds students that the 'puzzle' of why elements behave differently is beginning to have an answer.	Observable indicator: Each student's CER response contains all three components. Specifically check: (a) Claim names potassium as more reactive; (b) Evidence cites specific reactivity ratings from data cards; (c) Reasoning references the downward trend in Group 1, even if the atomic-level explanation (electron shells) is not yet present. Students who provide only a claim with no evidence are identified for one-on-one follow-up. Accept partial reasoning at this stage — full mechanistic explanation (valence electrons, shielding) is the target of Lessons 5–6.

	below chlorine, is less reactive. What does this tell us about the trend in Group 17?' Students revise or confirm their Phase 1 predictions in writing at the bottom of their Prediction Slip.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups retrieve the sticky-note questions they wrote at the end of Phase 2. The class DQB (a large wall display established in Lesson 1) is updated collaboratively. Step 1 (3 min): Each group posts their sticky note in one of four DQB zones — 'About Groups', 'About Periods', 'About Reactivity', or 'Still Confused'. Step 2 (5 min): Whole class reviews all new sticky notes. Teacher reads aloud 4–5 selected notes. Students vote (thumbs up) for the question they most want answered. Step 3 (2 min): Teacher highlights the two questions that will anchor the next two lessons: 'Why does reactivity increase down Group 1?' and 'Why does reactivity decrease down Group 17?' — and writes both in a special 'Next Lesson Will Answer' box on the DQB. Students copy both questions into their science notebooks under the heading 'DQB Update — Lesson 4'.	Stand at the DQB wall. Say: 'Post your group's sticky note in whichever zone fits best — you have 90 seconds.' After posting, read 4–5 notes aloud, including at least one from each zone. Say: 'Thumbs up for the question you most want answered this term.' Tally visible thumbs. Highlight the two anchor questions in a red marker box. Say: 'These two questions are sitting right at the heart of our driving question — why do elements behave so differently? The answer, which we will start building next lesson, is hidden inside the atom itself. Your element cards already carry a clue — the atomic number.' Point to atomic number on a displayed element card. Do not explain further. End with: 'Update your notebooks — Lesson 5 will start from exactly these two questions.'	Driving Question Board Collaborative Curation: The DQB functions as a public, cumulative record of the class's growing understanding — a physical artefact of the NGSS Science and Engineering Practice of Asking Questions and Defining Problems. Categorising questions into zones helps students recognise the difference between questions about patterns (observable) and questions about mechanisms (requiring deeper explanation), building metacognitive awareness of what they know versus what they still need to learn. The 'Next Lesson Will Answer' box creates forward momentum and intellectual anticipation.	Observable indicator: Every group has contributed at least one sticky-note question that goes beyond surface observation (e.g., not just 'What is Group 1 called?' but 'Why does potassium react more violently than lithium if they are in the same family?'). Teacher reviews all sticky notes after class to categorise the depth of student questioning. Questions that are purely definitional indicate groups that may need additional scaffolding on the distinction between pattern and mechanism in Lesson 5.
Model Building Phase 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table	Students return to their personal element grid (the annotated periodic table started in Lesson 1 and revised in Lessons 2–3). Model Revision Task (10 min): Using four different coloured pencils, students: (1) Shade Group 1 elements yellow and label the column 'Alkali Metals — reactivity ↑ going down'; (2) Shade Group 2 elements orange and label 'Alkaline Earth Metals'; (3) Shade Group 17 elements green and label 'Halogens — reactivity ↓	Distribute or instruct students to open their personal element grids. Say: 'We are upgrading our model with what we learned today. Use four colours — I will tell you what each colour means.' Read the five shading instructions aloud slowly, pausing after each for students to complete the shading before moving on. Circulate and check that trend arrows are drawn in the correct direction — Group 1 arrow should point DOWN for increasing reactivity; Group 17 arrow should	Cumulative Annotated Model Revision: Requiring students to physically update the same model across all 10 lessons embeds the NGSS Science and Engineering Practice of Developing and Using Models. The 'model limitation' sentence is a critical metacognitive move — it prevents students from treating the coloured grid as a complete explanation and keeps alive the productive tension that drives the storyline forward. The Kenyan context annotations	Observable indicator: Each student's element grid shows correct colour-coding for all five family regions, trend arrows pointing in the correct direction for Groups 1 and 17, at least two Kenyan context annotations, and a model limitation sentence that explicitly mentions the word 'why' or 'cause'. Grids missing the trend arrows or showing reversed arrows are collected for teacher review. The model limitation sentence is used as an exit-ticket

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	going down'; (4) Shade Group 18 elements blue and label 'Noble Gases — very low reactivity'; (5) Draw a bracket around the transition elements block and label it 'Transition Elements'. On their grid, students also add two Kenyan context annotations: a small drawing or label of a fertiliser bag next to potassium (K) with the note 'Rift Valley maize farms', and a water droplet next to chlorine (Cl) with the note 'Mombasa water treatment'. Finally, students draw directional trend arrows (↑ or ↓) along the left edge of Groups 1 and 17 labelled 'Reactivity trend'. Students write one 'model limitation' sentence at the bottom: 'My model shows WHERE trends occur but does not yet explain WHY they occur.'	point DOWN for DECREASING reactivity. If you spot a reversed arrow, ask: 'Wanjiku, which element in Group 17 is the most reactive — fluorine at the top or iodine at the bottom? So which direction does reactivity go?' (10 seconds WAIT TIME). After model revision, cold-call 2 students: 'Read me your model limitation sentence.' Affirm both. Close with: 'Your model now shows the families and the trends. Next lesson, we go inside the atom to explain the WHY — electron arrangement will be the missing piece.'	(fertiliser bag, water droplet) anchor abstract chemical families to lived experience, reinforcing the relevance of the periodic table to Kenyan agriculture and public health.	proxy — students who write 'My model is complete' (no limitation identified) need targeted questioning at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, review the following: (1) CER responses from Phase 3 — what percentage of students included all three components? If fewer than 70% included a reasoning component, plan a 5-minute whole-class modelling of the CER structure at the start of Lesson 5. (2) DQB sticky notes — are student questions mechanistic ('Why does…?') or purely descriptive ('What is…?')? A dominance of descriptive questions suggests students have not yet made the leap from pattern to mechanism and may need additional bridging at the start of Lesson 5. (3) Element grid models — are trend arrows consistently correct for both Group 1 and Group 17? Reversed Group 17 arrows are a strong predictor of misconception about halogen reactivity; address this explicitly in the Lesson 5 opening. (4) Consider: Did the Kenyan context examples (Rift Valley potassium fertilisers, Mombasa water treatment, Maasai cattle bones) generate genuine curiosity or were they treated as superficial decoration? If the latter, build in a 2-minute 'local connection' discussion at the start of Lesson 5 asking students to name one element family they encountered before school today.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students sorted Chemical Family Data Cards into Groups 1, 2, 17, 18, and transition elements; recorded reactivity and physical property trends in a Trend Detective Table; and identified that Group 1 reactivity increases going down (Li → Na → K) while Group 17 reactivity decreases going down (F → Cl → Br).
What did I learn?	Students learned the names and positions of the five major chemical families in the periodic table; that elements in the same group share similar properties due to their family membership; that reactivity trends within groups are directional and predictable; and that Kenyan materials such as potassium fertilisers (Rift Valley), chlorine for water treatment (Mombasa), and calcium in Maasai cattle bones are direct real-world expressions of these chemical families.
How does this explain the	Students explained, using Claim-Evidence-Reasoning, that potassium is more reactive than lithium because reactivity increases

phenomenon?	down Group 1; they identified that the position of an element in the periodic table predicts its behaviour; and they acknowledged that their current model shows WHERE the trends occur but cannot yet explain WHY — pointing to electron arrangement as the next piece of evidence needed to fully answer the driving question.

LESSON 5 (40 minutes): Ion Formation: Cations and Anions	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will predict whether elements lose or gain electrons to achieve stability, then construct dot-and-cross diagrams showing ion formation for selected elements from the first 20.
Knowledge	Students will know that atoms form cations by losing electrons and anions by gaining electrons; that the octet rule drives ion formation; and that an element's group number predicts how many electrons it loses or gains.
Skills	Students will draw dot-and-cross diagrams for Na, Cl, K, Ca, O, and S showing both the neutral atom and its resulting ion; write the charge and symbol of each ion formed; and use electron arrangement notation to justify predictions.
Attitudes	Students will demonstrate intellectual honesty by recording their pre-instruction predictions before receiving explanations, and openness to revising ideas when evidence contradicts initial thinking.
Key Inquiry Question	How do chemical bonds influence the properties of substances? This lesson provides the first mechanistic answer: before bonding can occur, atoms must first become ions by gaining or losing electrons, and the periodic table position tells us exactly which process each element undergoes.
Purpose in Storyline	In Lessons 3 and 4 students discovered that group position reflects outer-shell electron count and predicts reactivity trends. Now students go one level deeper: they see WHY sodium and chlorine behave as they do on the element cards — sodium loses one electron becoming Na+, chlorine gains one becoming Cl-. This invisible electron-transfer event is the hidden engine behind all the card-sort observations from Lesson 1, advancing the storyline from pattern to mechanism.
Safety Notes	No hazardous materials are used in this lesson. All demonstrations are conceptual and diagrammatic. Ensure pencils and rubbers are available for diagram work. No special PPE required.

B. LESSON OVERVIEW
Students begin by writing unaided predictions about electron loss or gain for Na, Mg, Cl, and O. The class shares predictions and areas of disagreement are marked as live questions. The teacher then demonstrates dot-and-cross diagrams for Na forming Na+ and Cl forming Cl- on the board, pausing for structured WAIT TIME at each step. Students immediately apply the technique to K, Ca, O, and S, working individually then checking in pairs. A targeted misconception-busting segment addresses the language of atoms wanting electrons, replacing it with stability and the octet framework. The lesson closes with an updated Driving Question Board entry and a revised element card annotation linking ion type back to the original card-sort phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Noble gas configuration]	Students open their notebooks to a fresh page headed PREDICTIONS — BEFORE INSTRUCTION. The teacher displays four element	Teacher says: LOOK at the four element cards on the board — you sorted these in Lesson 1 without knowing why they behave	Prediction journaling anchors prior knowledge and creates cognitive dissonance. By making predictions public (pair-share) before	Teacher circulates and reads 6 to 8 notebooks, noting: (a) whether students correctly recall electron arrangements from Lesson 3 — if

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	cards from the Lesson 1 card sort: sodium (Na, used in table salt that seasons ugali across Kenya), magnesium (Mg, found in green vegetables like sukuma wiki), chlorine (Cl, used in water treatment at Gigiri and Mombasa), and oxygen (O, present in air and water everywhere). For each element, students write three things without help: (1) the full electron arrangement using the notation from Lesson 3 (e.g., 2,8,1 for Na); (2) whether they predict the atom will LOSE or GAIN electrons; and (3) how many electrons they predict it will lose or gain. Students work in silence for 5 minutes. After writing, students share predictions with a neighbour and place a tick where they agree and a question mark where they disagree. The disagreement points are the productive tension that the lesson will resolve.	differently. Today we start answering that question at the electron level. Before I teach you anything, I want your best current thinking in your notebook. Teacher circulates silently during the 5-minute writing window, reading predictions without commenting, noting which elements produce the most disagreement (typically oxygen and magnesium). After the pair-share, teacher asks: Put your hand up if you and your partner disagreed about oxygen. Count hands visibly and say: GOOD — that disagreement is exactly the puzzle we are solving today. WAIT TIME: After asking any student to share a prediction, teacher waits a full 5 seconds before responding or calling on another student.	instruction, students invest in the outcome and pay closer attention to where the teacher-demonstration confirms or challenges their thinking. Linking to the original card-sort cards re-activates the anchoring phenomenon.	more than one-third have errors, a quick 2-minute whole-class correction is inserted before moving on; (b) which direction (lose vs. gain) students predict for each element — this data shapes which misconceptions to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Noble gas configuration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher draws a large dot-and-cross diagram for neutral sodium (Na) on the board step by step: nucleus labelled with atomic number 11, then inner shells 2 and 8, then the single outer-shell electron shown as a dot. Teacher then draws the resulting Na+ ion next to it: same nucleus, only two complete shells (2 and 8), no outer electron, and the 1+ charge written in a circle at the top right. Students observe and are NOT asked to copy yet — they are asked to WATCH. Teacher then repeats the process for neutral chlorine (Cl): nucleus 17, shells 2, 8, 7, with the single gap in the outer shell shown clearly. The Cl- ion is drawn beside it: nucleus 17, shells 2, 8, 8, with 1- charge. Only after both diagrams are complete does	Teacher narrates aloud while drawing, using precise language: The sodium atom has ONE electron in its outer shell. That outer shell wants to be full or empty for stability. Losing ONE electron costs less energy than gaining SEVEN, so sodium LOSES. The resulting ion has a full second shell — that is the stable configuration. WAIT TIME: After completing the Na+ diagram, teacher pauses for 8 seconds and says nothing, allowing students to process before moving to Cl. Teacher uses a coloured marker (red for electrons lost, blue for electrons gained) consistently throughout the demonstration so students can track direction of transfer visually. After students copy, teacher asks: In one	Observe-before-copy technique (teacher models completely before students reproduce) prevents students from writing and missing the conceptual narrative. Colour coding makes electron transfer direction a visible, observable feature rather than an abstract claim. Connecting to the Gigiri water-treatment photograph re-anchors the observable phenomenon to the invisible mechanism.	Teacher scans copied diagrams for three common errors: (1) incorrect number of electrons in each shell of the ion; (2) charge omitted or written incorrectly; (3) shells redrawn for the neutral atom rather than the ion. Teacher addresses errors immediately by saying: Check — your Na+ ion diagram should have exactly 10 electrons total. Count them now. Hands up if your count is not 10. This gives real-time data on readiness to proceed to student practice.

	teacher say: NOW copy both diagrams exactly into your notebook, labelling the charge and writing the ion symbol. Students copy for 3 minutes. Teacher points to the Nairobi Gigiri water-treatment photograph on the Cl card and says: The chlorine added to purify your drinking water exists as Cl- ions in solution — this diagram is showing you what is happening invisibly inside that water tank.	sentence, what is the difference between the Na atom diagram and the Na+ diagram? WAIT TIME of 5 seconds before taking responses. Accept 3 student responses and chart them on the side of the board.		
Explain Phase 📹 **VIDEO:** [Noble gas configuration] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Using the two teacher-drawn models as reference, the class constructs a shared explanation of the octet rule. Teacher writes the term OCTET RULE on the board and asks students to attempt a definition in their own words before giving one. After 3 student suggestions are heard, teacher provides: Atoms react in ways that result in a full outer shell of 8 electrons (or 2 for the first shell), because this arrangement is the most stable. Students then work individually to draw dot-and-cross diagrams for four new ions: K forming K+, Ca forming Ca2+, O forming O2-, and S forming S2-. They write (a) the neutral atom diagram, (b) the ion diagram, (c) the ion symbol with charge, and (d) one sentence stating whether the atom lost or gained electrons and why. Peer-check follows: partners swap books and use a three-point checklist (correct electron count in ion, correct charge symbol, correct ion notation). Misconception segment: Teacher addresses the language trap directly, saying: You may be tempted to write that sodium WANTS to lose an electron. Chemists do not use the word	Teacher circulates during individual diagram work (approximately 6 minutes), asking targeted questions to individual students: How many electrons does calcium have in its outer shell? So will it lose or gain electrons? How many? What charge does the ion carry? WAIT TIME: After each question, teacher waits 5 seconds before offering a prompt. If a student is stuck on Ca2+, teacher says: Look at the electron arrangement 2,8,8,2 — how many are in the outer shell? What is the quickest way to get to a full outer shell? When reviewing peer-checks, teacher asks: Did your partner have the correct electron count in the O2- ion — it should be 10. If not, coach them to find the error before moving on. For the misconception segment, teacher uses a specific quote: Scientists discovered that atoms do not choose anything — the laws of physics make certain configurations more stable. Mendeleev and Moseley did not know about stability the way we do now — that understanding came later.	Gradual release: teacher models two cases fully (Na, Cl), students independently produce four cases (K, Ca, O, S), building procedural fluency. Peer-checking with a specific checklist makes self-correction a social and structured process rather than vague comparison. The misconception-busting language exercise develops scientific precision and models how real chemists think — connecting to the storyline goal of students thinking like the historical chemists on the DQB timeline.	Peer-check checklist provides immediate feedback on three key accuracy points. Teacher collects 5 random notebooks after the pair-check and quickly reviews Ca2+ and S2- diagrams — these are the two highest-error items. Results inform whether the Model Building phase needs additional scaffolding or can proceed at pace. Teacher also listens for persistence of the word want in student verbal explanations during the misconception segment.

	want for atoms — atoms are not alive and do not have desires. We say that losing one electron results in a MORE STABLE arrangement. Rewrite any sentence in your notebook that uses the word want and replace it with a stability-based explanation. Students revise language in their notebooks.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Noble gas configuration] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students return to the Driving Question Board posted at the front of the room. From Lesson 4, two open questions are displayed: Why does reactivity increase down Group 1 but decrease down Group 17? and How does the position of an element in the periodic table tell us how it will react? Each student receives one sticky note. They write a REVISED or NEW question, or a partial answer to an existing question, that today's lesson now allows them to address. Examples that may emerge: Now I know Na loses 1 electron because it is in Group 1 — does every Group 1 element form a 1+ ion? Why does fluorine form F- and not F+? Does the ion an element forms tell us what compound it will make? Students post their notes on the DQB in two zones: ANSWERED (questions today cleared up) and NEW WONDERS (questions today opened up). The teacher reads 4 to 5 notes aloud and says: Notice how answering one question always opens more questions — that is exactly what happened to Mendeleev when he predicted missing elements.	Teacher points to the element card sort photographs from Lesson 1 displayed beside the DQB and says: When you first sorted these cards, you grouped sodium and potassium together because they looked similar on the surface. Today we can say WHY at the electron level — both lose exactly one electron to form 1+ ions. That invisible event is what makes them behave similarly on the outside. Teacher reads the Lesson 4 question about Group 17 reactivity aloud and says: Can anyone post a partial answer using today's ion diagrams as evidence? Accept 2 student responses and acknowledge them by posting the sticky note themselves to model the practice. WAIT TIME: After asking for new questions to post, teacher waits 10 seconds in silence before prompting — this silence signals that thinking is expected, not optional.	DQB updates make learning visible and cumulative, showing students that the storyline is progressing deliberately. Connecting the updated board back to the Lesson 1 card sort maintains the anchoring phenomenon as the reference point for all new learning. The Mendeleev parallel reinforces the science-as-a-process narrative established in Lesson 2.	Quality of sticky notes reveals depth of understanding: surface-level notes (e.g., I learned ions have charges) indicate procedural uptake only; deeper notes (e.g., if Group 2 elements lose 2 electrons to form 2+ ions, does that mean they make different compounds from Group 1?) indicate conceptual transfer. Teacher photographs the DQB after each lesson for tracking purposes.
Model Building Phase 📹 **VIDEO:**	Students retrieve their element cards from Lesson 1 (or a printed copy). Using the ion diagrams drawn in the Explain phase, they	Teacher circulates and asks students to read their generalisation sentences aloud in a low-voice sharing format	Re-annotating the original element cards directly connects the new mechanistic understanding (ion formation) back to the anchoring	Teacher collects the four generalisation sentences (written on a half-sheet or in notebooks) as an exit task. Success criteria:

[Noble gas configuration](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	add a new annotation to the card for Na, Cl, K, and Ca: a small box in the corner of the card labelled ION FORMED, containing the ion symbol (e.g., Na+), a mini dot-and-cross of the ion, and one Kenyan context sentence. Example annotation for Na: ION FORMED — Na+. Sodium loses 1 electron from its outer shell. Na+ ions are found dissolved in the Malindi sea water used to make coastal sea salt. Example for Ca: ION FORMED — Ca2+. Calcium loses 2 electrons. Ca2+ ions form part of the bone structure of Maasai cattle in the Rift Valley. Students then place the four annotated cards back into their periodic table grid from Lesson 3 and answer one synthesis question in their notebook: Looking at the grid, what pattern do you see between an element's group number and the ion it forms? Students write a generalisation statement: Group 1 elements form ___ ions with a charge of ___. Group 2 elements form ___ ions with a charge of ___. Group 16 elements form ___ ions with a charge of ___. Group 17 elements form ___ ions with a charge of ___. This generalisation becomes the bridge to Lesson 6, where valency and oxidation numbers are formalised.	(students read to their immediate neighbour simultaneously, then one group is invited to share with the class). Teacher writes the four generalisation statements on the board as students contribute them, then says: Hold this idea — in our next lesson we give these patterns a formal name: VALENCY. And we will use valency to do something powerful — figure out the formula of any compound without memorising it. Teacher points to the Na card and says: The mystery of why sodium on your original card-sort behaved so differently from chlorine, even though they sit next to each other in Period 3, is now solved at the electron level. Sodium is racing to lose one electron; chlorine is racing to gain one. They are going in opposite directions — and that is why they are so different. WAIT TIME: After asking for the generalisation about Group 16, teacher waits 8 seconds without offering the answer.	phenomenon (card sort) and makes the model revision visible and tangible. The generalisation exercise begins the process of abstracting a rule from specific cases, preparing students for the formal valency table in Lesson 6. The spatial act of replacing cards into the periodic table grid reinforces that position and electron behaviour are linked.	correct ion symbol, correct charge, correct group association. This data is used to group students for targeted support in Lesson 6 — students who cannot yet write the Group 16 or Group 17 generalisation correctly will receive a brief scaffolding card at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, consider: Did students predictions before instruction reveal any systematic misconceptions about which direction elements transfer electrons — especially for Group 16 and 17 elements? Were students able to draw dot-and-cross diagrams for Ca2+ and S2- independently, or did they need prompting for multi-electron transfers? Did the misconception-busting segment (replacing want with stability language) take root — were students still using anthropomorphic language at the end? Did the DQB sticky notes show surface-level or deep conceptual thinking? Were Kenyan contexts (Malindi salt, Maasai cattle bones, Gigiri water treatment, sukuma wiki magnesium) mentioned spontaneously by students or only when prompted by the teacher? Which students are ready to move directly into valency formalisation in Lesson 6, and which need the scaffolding card?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	When sodium and chlorine element cards were placed next to each other in our periodic table grid, they showed completely different reactivity ratings and contexts — sodium in table salt and chlorine as a disinfectant gas. We observed from dot-and-cross diagrams on the board that sodium ends up with 10 electrons after losing one, and chlorine ends up with 18 electrons after gaining one.
What did I learn?	Atoms form ions by losing electrons (forming positively charged cations) or gaining electrons (forming negatively charged anions). The number of electrons lost or gained is determined by the element's group number and the drive toward a full outer shell described by the octet rule. Na forms Na^+, K forms K^+, Ca forms Ca^{2+}, Cl forms Cl^-, O forms O^{2-}, and S forms S^{2-}.
How does this explain the phenomenon?	The element card for sodium showed high reactivity and a 1+ charge tendency, while chlorine showed a tendency to react as a 1- species — today we explained this using electron arrangement. Sodium is in Group 1 with one outer-shell electron and loses it to achieve the stable 2,8 configuration. Chlorine is in Group 17 with seven outer-shell electrons and gains one to achieve the stable 2,8,8 configuration. Their opposite electron-transfer directions explain why elements sitting next to each other in Period 3 behave so completely differently — answering part of the Lesson 1 puzzle from the card sort.

LESSON 6 (80 minutes): Oxidation Numbers and Valency — Reading the Periodic Table's Charge Language	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students infer valency and oxidation numbers directly from electron arrangements, tabulate group trends for Groups 1, 2, 16, and 17, and extend thinking to variable oxidation numbers in transition elements — building the conceptual bridge between invisible electron structure and visible chemical behaviour.
Knowledge	Students will know that: (i) valency is the combining power of an element determined by its number of valence electrons or electron vacancies; (ii) oxidation number represents the charge an atom carries (or appears to carry) when bonding; (iii) Groups 1, 2, 16, and 17 have fixed oxidation numbers (+1, +2, −2, −1 respectively) predictable from periodic table position; (iv) transition elements such as iron exhibit variable oxidation numbers (Fe^{2+} and Fe^{3+}); (v) radicals are groups of atoms that carry a net charge and behave as a single unit in chemical reactions (SO_4^{2-}, NO_3^-, CO_3^{2-}, OH^-, NH_4^+).
Skills	Students will be able to: (i) infer valency and oxidation numbers from electron arrangements of the first 20 elements; (ii) tabulate valencies for Groups 1, 2, 16, and 17 elements; (iii) identify and recall the charges and valencies of common radicals; (iv) discuss elements with variable oxidation numbers using evidence from iron compounds; (v) write electron arrangements of selected ions using s and p notation.
Attitudes	Students appreciate that understanding oxidation numbers unlocks the ability to predict how substances form — showing compassion and tolerance while discussing with peers how atoms acquire stability (Peace value). Students value the contributions of diverse scientists to our understanding of ion formation (Citizenship competency).
Key Inquiry Question	How do chemical bonds influence the properties of substances?
Purpose in Storyline	In Lesson 3, students mapped electron arrangements for H–Ca onto the periodic table. In Lesson 5, students gathered evidence about ion formation (cations and anions). Lesson 6 is the Explain & Model pivot: students now infer valency and oxidation numbers from those electron maps, quantify the 'charge language' each group speaks, and encounter the puzzle of variable oxidation numbers in transition metals — concretely linked to the rusting railings at Kisumu Port. This lesson directly advances the Driving Question by showing that an element's position encodes its oxidation number, giving students a predictive tool for the Mystery Element Card Challenge. The DQB is updated to carry forward the new question: 'How do we know which oxidation number to use?'
Safety Notes	No hazardous chemicals are used in this lesson. All activities are paper/whiteboard based. If iron(II) chloride or iron(III) chloride samples are displayed for observation, students must not taste or directly handle the salts; teacher handles sample vials. Wash hands after any contact with demonstration materials.

B. LESSON OVERVIEW
Lesson 6 is a fully evidence-rich Explain & Model lesson in which students move from observed electron arrangements (Lessons 3–5) to inferring valency and oxidation numbers. The lesson opens with a Predict task connecting back to the Mystery Element Cards: students predict the 'charge' a sodium atom must carry to become stable, then build outward to Group trends. A structured data-table activity (Observe) has students tabulate valencies for Groups 1, 2, 16, and 17 from their own Bohr diagrams. The Explain phase introduces oxidation numbers formally, uses rust at Kisumu Port as the anchor for variable oxidation numbers (Fe^{2+}/Fe^{3+}), and introduces radicals through Kenyan chemical contexts (sulfate in fertiliser used in Rift Valley farms, carbonate in limestone from Bamburi Cement, nitrate in Kenyan horticultural exports, hydroxide in soap-making, ammonium in DAP fertiliser). A Driving Question Board update consolidates new vocabulary, and the lesson closes with a cumulative particle model revision in which students annotate their element cards with oxidation numbers for the first time.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Simplifying Radicals Review Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students receive their Mystery Element Cards from Lesson 1 (or a class set displayed on the board). Working individually for 3 minutes, students write a prediction in their notebooks: 'If sodium (Na, atomic number 11) must reach the electron arrangement of neon (2,8) to become stable, what charge must the sodium ion carry? Write your prediction and your reasoning.' After writing, students share predictions with a shoulder partner (Think-Pair-Share) for 2 minutes. The class then views two photographs on screen: (1) a bag of table salt (sodium chloride) used to season ugali at a Nairobi market; (2) a jar of sodium hydroxide used by a soap-maker in Kisumu. The teacher asks: 'Both photos show sodium behaving the same way — always combining with one other ion. What does that tell you about sodium's charge?' Students record a refined prediction. The teacher then extends: 'Look at your Mystery Element Card for magnesium and calcium — predict their charges using the same logic.' Students write a second prediction for Mg and Ca. A brief whole-class share (cold-call, 3 students) captures the range of predictions before any instruction is given.	1. Distribute or display Mystery Element Cards. Say: 'Before I teach you anything today, I want you to be a detective. Look at sodium's electron arrangement — 2, 8, 1 — and think about what we discovered last lesson about ions.' WAIT 10 seconds. 2. Issue the written prediction prompt. Circulate silently for 3 minutes — do not confirm or deny. Note misconceptions on a sticky note for later targeting. 3. Launch Think-Pair-Share: 'Turn to your partner. Share your prediction. Your partner must say whether they agree or disagree — and why.' WAIT 2 minutes. 4. Display the ugali salt photograph. Say: 'Sodium always does the same thing in every compound. Why? What is its electron arrangement telling it to do?' Cold-call 3 students by name. Probe the third with: 'You said +1 — how many electrons did sodium lose to get there? Show me on your fingers.' 5. Extend to Mg and Ca: 'Now use the same logic for the alkaline earth metals on your cards. Write your prediction for the charge of Mg^{2+} and Ca^{2+} before I confirm anything.' WAIT 15 seconds. 6. Record all class predictions visibly on the board under the heading 'Our Predictions — we will test these.'	Think-Pair-Share with Evidence Anchor: Students first commit to a written prediction (reducing passive reception), then stress-test it against a partner's reasoning, then anchor it to a visible Kenyan photograph. This sequence activates prior knowledge from Lesson 3 (electron arrangements) and Lesson 5 (ion formation) and creates productive cognitive dissonance when predictions for Mg and Ca vary across the class — generating genuine need-to-know before the data table activity.	Observable indicator: Students' written predictions correctly identify Na^{+} (loss of 1 electron → +1 charge) and at least attempt Mg^{2+}/Ca^{2+}. Misconception to watch: students who write Na^{2+} or Na^{-} — these students have not yet connected valence electron count to ion charge direction. Teacher notes these names for targeted cold-calling in Phase 2. Exit check: teacher scans notebooks during circulation — at least 70% of students should show charge = group number logic before Phase 2 begins.
Observe Phase 📹 **VIDEO:** Simplifying Radicals Review Principles - Basic	Students work in groups of four. Each group receives a printed Group Valency Data Table (blank) with columns: Element | Symbol | Atomic Number | Electron Arrangement | Valence Electrons |	1. Distribute blank data tables. Say: 'You already know the electron arrangements from Lesson 3. Your job now is to find the pattern between the group number and the charge. Do not	Structured Data Table with Reconciliation Protocol: The blank-table-then-self-correct sequence (rather than a pre-filled table) forces students to generate knowledge from prior evidence	Observable indicator: Completed data tables show correct oxidation numbers for all 12 elements with fewer than 2 errors per group after self-correction. Specific check: teacher collects one table per

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Charge of Ion | Oxidation Number | Group Number. Groups must fill in the table for 12 elements: Li, Na, K (Group 1); Mg, Ca (Group 2); O, S (Group 16); F, Cl, Br (Group 17); and Ne, Ar (Group 18, noble gases — as a reference baseline). Students draw on their Lesson 3 electron arrangement notes and their Mystery Element Cards. A projected reference strip shows the first 20 elements in order (atomic number only, no charges). After 8 minutes of group table completion, groups compare tables with the adjacent group and reconcile any differences (2 minutes). The teacher then reveals a completed reference table on screen one row at a time, pausing after each group for students to self-correct. Students highlight any cell they changed in a different colour — these become their 'revision flags' for the end of lesson. To anchor the observation in the phenomenon, students are shown a short photographic sequence: chlorine gas used at Nairobi's Gigiri water treatment plant → the chloride ion in table salt → fluoride in Kenyan toothpaste brands → the pattern: 'All Group 17 elements carry a −1 charge.' Students annotate this pattern directly onto their Mystery Element Cards for the first time.	guess — use your data.' WAIT 10 seconds. 2. Circulate to groups. After 4 minutes, prompt any group that is stuck with: 'How many electrons does fluorine need to reach neon's arrangement? That tells you the charge.' Do NOT give the answer — ask the question and move on. 3. After 8 minutes, say: 'Stop writing. Compare your Group 17 row with the group next to you. If you disagree, you have 2 minutes to argue your case using the electron arrangement — not just your feeling.' 4. Reveal reference table one group at a time on the projector. After Group 1: 'What is the pattern? Finish this sentence in your book: Group 1 elements always have an oxidation number of ___.' Cold-call 2 students. After Group 17: 'Why is the charge negative for Group 17 but positive for Group 1? Explain using electrons, not just the rule.' Cold-call 3 students — specifically call on students who had revision flags. 5. Display the Gigiri water treatment photo. Say: 'This is chlorine doing its job to make Nairobi's water safe. When chlorine becomes an ion in that water, what charge does it carry? How does your table prove that?' WAIT 15 seconds. Cold-call 1 student.	rather than copy it. The colour-coded revision flags make metacognition visible — students literally see where their mental model was incomplete. Connecting each group to a Kenyan photograph (Gigiri plant, Rift Valley fertiliser, Bamburi limestone) keeps the data anchored to real phenomena rather than abstract numbers.	group at end of phase and scans for the noble gas row — students who assign a charge to Ne or Ar (rather than zero/stable) have a fundamental misconception about full outer shells that must be addressed in Phase 3. Listen for the phrase 'valence electrons' during group discussions — students who use 'electrons' generically rather than 'valence electrons' need a vocabulary correction before Phase 3.
Explain Phase 📹 **VIDEO:** Simplifying Radicals Review Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Part A — Fixed Oxidation Numbers (8 min): Teacher formalises the pattern students found in Phase 2. Students write the definition of oxidation number in their own words, then compare with a partner, then hear the formal KICD definition. Students complete a quick application: given three Mystery Element Cards (one	Part A: 1. Write on board: 'Oxidation number = the charge an atom carries when it forms an ion.' Say: 'Write this in your own words first — do not copy mine.' WAIT 20 seconds. Cold-call 2 students to read their version. Say: 'Good — now let's see if your version works when we apply it.' 2. Demonstrate the crossover valency method for	Three-part nested explanation with progressive complexity: Part A consolidates fixed patterns (deductive from data), Part B introduces productive anomaly (variable oxidation numbers — inductive from evidence about iron), and Part C extends the framework to polyatomic units (radicals). The Kenyan context	Observable indicators: (1) Students correctly derive Fe^{2+} and Fe^{3+} from charge balance without being told — teacher listens for the calculation process, not just the answer. (2) Radicals Reference Cards are completed with correct charges for all five radicals — teacher stamps only complete, accurate cards. (3) Exit

Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Group 1, one Group 2, one Group 17), write the formula of the compound each element forms with oxygen — using the crossover valency method demonstrated by the teacher on the board (e.g., Na_2O, MgO, Cl_2O — noting that Cl_2O is unusual and previewing that Cl has variable oxidation numbers in some compounds). Part B — Variable Oxidation Numbers: The Kisumu Port Rust Story (10 min): Teacher displays two photographs side by side: (i) iron railings at Kisumu Port showing orange-brown rust; (ii) a clean iron nail. Teacher says: 'Same element, iron — but the rust and the nail behave differently. Let's find out why.' Students are shown molecular models/diagrams of $FeCl_2$ (iron(II) chloride, pale green) and $FeCl_3$ (iron(III) chloride, dark brown/orange). Using their data table logic, students work in pairs to answer: 'If each chloride ion is −1, what must the iron charge be in each compound to make the formula electrically neutral?' After 3 minutes, teacher confirms Fe^{2+} and Fe^{3+} and introduces the term 'variable oxidation numbers.' Students annotate the Kisumu Port photographs in their notebooks: 'Fe^{2+} in green rust, Fe^{3+} in orange-brown rust (common form).' Teacher explains that iron's variable oxidation number is why the colour of rust depends on conditions — a real-world pattern connected to the periodic table. Part C — Radicals (7 min): Teacher introduces five radicals using a Kenyan context anchor for each: SO_4^{2-} (sulfate — in CAN fertiliser used on Rift Valley tea farms), NO_3^- (nitrate — in KNO_3	Na_2O on the board, narrating each step aloud. Then say: 'Now you try MgO using the same method. Work alone — 90 seconds.' WAIT. Cold-call 1 student to narrate their steps aloud. Part B: 3. Display the Kisumu Port photographs. Say: 'Iron. Same element. Different compounds. Your job is to use charge balance — like a financial account that must add to zero — to find iron's charge in each compound.' WAIT 10 seconds. 4. During pair work, circulate and listen for students using the phrase 'neutral compound.' If a group is struggling, ask: 'If you have two chloride ions, each −1, what is the total negative charge? What positive charge cancels that out?' Do NOT say Fe^{2+} — let them derive it. 5. After 3 minutes: cold-call 3 different students — one for $FeCl_2$, one for $FeCl_3$, one to explain in their own words why the same element can have two different oxidation numbers. 6. Say: 'Write in your notebook: The rusting of iron railings at Kisumu Port produces Fe^{3+} ions. This is why rust is orange-brown — the oxidation number of iron changes depending on conditions.' Part C: 7. For each radical, point to the Kenyan photograph/context on the projector and say: 'This radical is already in your world — you've seen it, eaten it, or used it. Now we're naming the charge it carries.' 8. Give students 5 minutes to complete the Radicals Reference Card independently. Circulate and stamp/initial completed cards. Say: 'Keep this card — you will need it in every lesson from now on.' 9. Close Part C with: 'Ammonium is special — it is the only positively-charged radical on your card.	anchor for each radical (Rift Valley fertiliser, Bamburi limestone, Bidco soap, Naivasha horticulture, DAP fertiliser) ensures each new abstract symbol is tied to a concrete, locally familiar substance — reducing cognitive load and increasing retention. The charge-balance framing ('like a financial account that must add to zero') gives students a transferable mental model for all future formula work.	misconception probe: teacher asks 'What is the valency of SO_4^{2-}?' — correct answer is 2 (matching the magnitude of the charge). Students who say 'negative 2' instead of '2' understand the charge but not the valency distinction — note for targeted follow-up in Lesson 7.

	used in Kenyan horticultural export crops, Naivasha), CO_3^{2-} (carbonate — in limestone from Bamburi Cement, Mombasa), OH^- (hydroxide — in NaOH used by Bidco soap factory, Thika), NH_4^+ (ammonium — in DAP fertiliser used across Western Kenya). Students complete a Radicals Reference Card (structured worksheet) filling in: name, formula, charge, valency, and one Kenyan use. This card is kept in their exercise books as a reference tool for Lesson 7 (formula writing).	Every other radical is negative. Why might that matter when we write formulas?' WAIT 15 seconds. Cold-call 1 student.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Simplifying Radicals Review Principles - Basic] Source: CK-12 [🔍 Search ARES for similar videos] 📖 **HTML:** [Observation: Penny Oxidation (PLIX)] Source: CK-12 [🔍 Search ARES for similar readings]	Students return to the class Driving Question Board. The teacher reads aloud the driving question: 'Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?' Students individually write on a sticky note (or digital card on the ARES platform): one thing they can now answer about the driving question, and one new question that today's lesson raised. Suggested new question (teacher models if students are stuck): 'How do we know which oxidation number to use when a transition element like iron has more than one option?' This question is written on the DQB in a new colour (e.g., orange) to mark it as a 'Lesson 6 question.' The teacher also adds a 'We now know' strip to the board: 'An element's GROUP NUMBER tells us its fixed oxidation number. Transition elements have VARIABLE oxidation numbers. RADICALS carry charges and act as a single unit.' Students update their personal DQB tracking sheet	1. Point to the driving question displayed on the board. Say: 'Six lessons in. Let's audit what we know.' Give students 3 minutes to write their sticky note — one answer, one new question. Do not prompt content yet. 2. Collect 5–6 sticky notes at random. Read them aloud without naming the author. After each: 'Does the class agree this is answered? Thumbs up, thumbs sideways, thumbs down.' 3. Display the new orange DQB question: 'How do we know which oxidation number to use?' Say: 'This is our new unresolved question. We will carry it into Lesson 7 when we write formulas — and you'll need to choose the right oxidation number.' 4. Add the 'We now know' strip. Say: 'Copy these three statements into your DQB tracking sheet — they are your evidence that the driving question is becoming answerable.' 5. Run the class vote on which prior DQB question feels most answered: 'Raise your hand for the question you think we've made the most progress on today.' Tally votes visibly. Celebrate progress:	Metacognitive DQB Audit with Voting Protocol: By explicitly tracking 'what we now know' versus 'what we still don't know,' students develop Learning-to-Learn competency (KICD core competency). The voting protocol makes collective understanding visible and gives quieter students a voice. Colour-coding DQB questions by lesson number allows students to trace the narrative arc of the unit — building the sense that science is a cumulative, evidence-driven story rather than a collection of isolated facts.	Observable indicator: Student sticky notes correctly identify at least one of the three 'We now know' statements as an answer to the driving question, and pose a genuine new question (not simply restating today's content). Red flag: students whose 'new question' is blank or trivial (e.g., 'What is an element?') — these students may not yet see the driving question as personally relevant; teacher follows up individually at the start of Lesson 7.

	(kept in exercise books) with these three statements. The class briefly reviews all DQB questions accumulated across Lessons 1–6 and votes (show of hands) on which question feels closest to being answered — building metacognitive awareness of the unit's progress.	'Look how far we've come from Lesson 1 when you couldn't explain why sodium and potassium behave the same way.'		
Model Building Phase 📹 **VIDEO:** Simplifying Radicals Review Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Mystery Element Card set (or the class display version). Working individually, students annotate five of their element cards (teacher assigns: Na, Mg, Cl, Fe, and one element of the student's own choice from H–Ca) with the following new information in a designated 'Lesson 6 Update' box on each card: (i) the oxidation number of the element's common ion, (ii) the ion symbol written in s and p notation (e.g., Na^+: $1s^2\ 2s^2\ 2p^6$), and (iii) one Kenyan context in which that ion appears. For iron, students write BOTH Fe^{2+} and Fe^{3+} and annotate: 'variable — see Kisumu Port rust.' Students then hold their annotated Fe card next to their Na card and respond in writing to the prompt: 'Na always has oxidation number +1. Fe can have +2 or +3. What does this difference tell you about where these elements sit in the periodic table?' This written response is the cumulative model artefact for Lesson 6 — it will be revisited in Lesson 9 (bonding and formulae) when students write the formulae of iron compounds from first principles.	1. Say: 'Your Mystery Element Cards are your growing model of the periodic table. Every lesson, you add a new layer of understanding. Today's layer is oxidation number.' Distribute cards or direct students to the display. 2. Give students 7 minutes to annotate the five cards. Circulate — check that ion symbols in s/p notation are correct (Na^+ should be $1s^2\ 2s^2\ 2p^6$, not $1s^2\ 2s^2\ 2p^6\ 3s^1$). If a student writes the electron arrangement of the neutral atom instead of the ion, ask: 'Na^+ has lost one electron — where did it come from? So what's left?' WAIT 10 seconds. Do not correct directly — use questioning. 3. After annotation, pose the comparison prompt aloud: 'Look at Na and Fe side by side. Na is in Group 1. Fe is a transition metal. Why does Fe get to have TWO different oxidation numbers when Na only gets one?' WAIT 15 seconds. Cold-call 2 students. 4. Collect the written responses as an exit artefact — scan quickly to check that at least 80% of students connect Fe's variability to its position as a transition element. 5. Close the lesson: 'Next lesson, we use today's oxidation numbers to write the formulae of real compounds — including compounds found in Kenyan homes, farms, and factories. Bring	Cumulative Annotated Model with Comparison Prompt: The Mystery Element Card annotation is the lesson's capstone sensemaking move — it forces students to synthesise all five phases into a single physical artefact. The Na vs. Fe comparison prompt is deliberately contrastive (fixed vs. variable oxidation numbers), prompting relational thinking rather than recall. Writing in s and p notation for the ion (not the neutral atom) consolidates the conceptual shift from atom to ion that underpins all of Sub-Strand 1.3.	Observable indicator: Annotated cards show correct ion symbols in s and p notation AND correct oxidation numbers for all five elements. Specific check: the Fe card must show both Fe^{2+} and Fe^{3+} — cards showing only one are incomplete. Written comparison responses that explain Fe's variability using 'transition metal' or 'd-block' language demonstrate above-expectation understanding. Written responses that say only 'Fe is different' without referencing periodic table position indicate the student has learned the fact but not the principle — teacher notes these for targeted questioning in Lesson 7.

		your Radicals Reference Card.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did students successfully derive Fe^{2+} and Fe^{3+} from charge balance, or did they rely on being told? If the latter, the charge-balance framing needs to be made more explicit in Lesson 7's formula-writing activities. (2) How many Radicals Reference Cards were fully and correctly completed? Cards with errors in SO_4^{2-} (commonly written as SO_4^{-} by students who confuse charge with atom count) signal a need for a 2-minute revision at the start of Lesson 7. (3) Were DQB sticky notes substantive? If students struggled to pose a new question, the driving question may need to be made more personally relevant — consider asking students to bring in a Kenyan product label from home (fertiliser bag, soap packet, cement bag) and identify one radical or ion on the label for Lesson 7. (4) Did the Kisumu Port rust context land as a genuine puzzle, or did it feel decorative? If students did not express surprise at Fe having two oxidation numbers, revisit the photograph at the start of Lesson 7 and ask: 'Why is some rust orange and some rust green?' — the colour difference is the perceptual hook that makes variability feel real. (5) Review the annotated Mystery Element Cards: the ratio of students who correctly wrote ion electron arrangements in s/p notation (vs. neutral atom) is your single best indicator of conceptual readiness for Lesson 7's formula-writing work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students tabulated valencies and oxidation numbers for Groups 1, 2, 16, and 17 from electron arrangement data; examined photographs of $FeCl_2$ and $FeCl_3$ alongside images of rusted and clean iron at Kisumu Port; and completed a Radicals Reference Card linking SO_4^{2-}, NO_3^{-}, CO_3^{2-}, OH^{-}, and NH_4^{+} to Kenyan contexts (Rift Valley fertiliser, Naivasha horticulture, Bamburi limestone, Bidco soap, DAP fertiliser).
What did I learn?	An element's group number in the periodic table predicts its fixed oxidation number (Group 1 → +1; Group 2 → +2; Group 16 → −2; Group 17 → −1). Transition elements such as iron have variable oxidation numbers (Fe^{2+} and Fe^{3+}), which is why rusted iron at Kisumu Port can appear different colours. Radicals are groups of atoms that carry a net charge and act as a single unit in chemical reactions; their valency equals the magnitude of their charge.
How does this explain the phenomenon?	The oxidation number of an element arises directly from the number of electrons it gains or loses to achieve a stable noble gas electron arrangement — connecting today's lesson back to the electron arrangement work in Lesson 3 and the ion formation evidence in Lesson 5. Transition elements have incompletely filled d sub-shells that allow them to lose different numbers of electrons depending on conditions, explaining variable oxidation numbers. This means that an element's position in the periodic table encodes its charge language — a direct, partial answer to the driving question.

LESSON 7 (40 minutes): Deriving Chemical Formulae Using Valency and the Crossover Method	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students use the valency and oxidation-number patterns they discovered from periodic table position to derive the correct chemical formulae of ionic compounds, making the invisible logic of element behaviour visible as a written formula.
Knowledge	Students will be able to state the valencies of given cations and anions, apply the crossover method to derive formulae of binary ionic compounds and compounds containing radicals, and write the correct formula with subscripts and brackets where needed.
Skills	Students will practise deriving formulae systematically, self-marking their work against an answer key, and identifying and naming their own error patterns in writing.
Attitudes	Students develop confidence in using a logical rule-based method, and appreciate that chemical formulae are not arbitrary but are determined by the periodic table structure they have already mapped.
Key Inquiry Question	How do chemical bonds influence the properties of substances? — Deriving formulae from valency reveals the fixed combining ratios that ionic bonds impose, directly answering why ionic compounds always appear in predictable proportions.
Purpose in Storyline	In Lessons 3 and 5 students mapped electron arrangements and ion charges. In Lesson 6 they built a valency table and met radicals. Lesson 7 is the payoff: all that invisible structure (electron shells, ion charges, oxidation numbers) now generates a concrete, testable product — the chemical formula. This prepares students for Lesson 8 where formulae are used to write and balance equations.
Safety Notes	No hazardous materials are used. This is a pencil-and-paper activity. Ensure students with visual impairments have enlarged tables.

B. LESSON OVERVIEW

Students open by predicting the formulae of two familiar Kenyan compounds before any instruction, surfacing prior reasoning. They then observe a teacher-guided worked example of the crossover method applied to calcium chloride (used in Kericho tea-farm soil conditioning). In the Explain phase the teacher unpacks the method step by step for four increasing-complexity examples, including radicals. Students practise eight formulae independently, self-mark against a projected answer key, and write an error-pattern sentence. The lesson closes with a DQB update and a revised element-card model showing formula beneath the ion diagram.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Second partial derivative test] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]	Students open their notebooks to the valency table they built in Lesson 6. The teacher projects two unlabelled element cards from the original Mystery Element Card Challenge: Card 11 (sodium) and Card 17 (chlorine), shown with their Kenyan photographs (ugali	Teacher says: Before I show you any method, I want to see your instinct. Look at your valency table from yesterday and use it as your only tool. Write your best prediction — there is no penalty for being wrong here, I just want to see your thinking. [WAIT TIME: 3	Prediction journaling activates prior knowledge of ion charges and valency from Lesson 6 and creates cognitive dissonance for students who predict NaCl2 or CaCl — wrong ratios they will self-correct during the Observe phase,	Teacher scans notebooks during circulation and mentally notes the most common incorrect ratio for calcium chloride (likely CaCl or CaCl2 written with uncertainty). This informs pacing: if most students already predict CaCl2 correctly, the worked example can

📖 **HTML:** [Observation: Penny Oxidation (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	salt and Gigiri water-treatment plant). Without any instruction on formula-writing, students write a prediction in their notebooks: What is the simplest formula you think represents one sodium combining with chlorine? Write the formula and explain in one sentence why you chose that ratio. Students have 3 minutes, working individually in silence. A second prompt is then shown: Card 20 (calcium) and Card 17 (chlorine) — predict the formula for calcium chloride, used to condition acidic soils on Kericho tea farms. Students write a second prediction with a ratio justification.	minutes of complete silence. Teacher circulates and reads notebooks without commenting.] After 3 minutes teacher says: Keep your predictions exactly as written — do not erase. We will return to them in a few minutes to see how close your instinct was to the rule.	making the correction memorable.	move faster; if NaCl2 appears frequently, the teacher will address the direction of the crossover explicitly.
Observe Phase 📹 **VIDEO:** [Second partial derivative test](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Observation: Penny Oxidation (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	Teacher draws a two-column crossover diagram on the board for calcium chloride, narrating each step aloud while students watch without writing. Step 1: Write the symbols with the ion charge above each: Ca^{2+} and Cl^{-}. Step 2: Cross the numbers only (not the signs): the 2 from Ca crosses to become the subscript of Cl; the 1 from Cl crosses to become the subscript of Ca, giving $CaCl_2$. Step 3: Simplify if the subscripts share a common factor (they do not here). Step 4: Check by verifying total positive charge equals total negative charge: 1 x (+2) = 2+ and 2 x (−1) = 2−. Balanced. Teacher then repeats the process for sodium chloride using the same four steps: Na^{+} and Cl^{-} cross to give Na_1Cl_1, simplified to NaCl. Students compare the observed NaCl and $CaCl_2$ against their predictions and circle whether their prediction was correct, partially correct, or needs revision.	Teacher says: Watch the whole example first — do not write yet, just observe the logic. [WAIT TIME: allows 60 seconds of silent observation after drawing the diagram before speaking.] Teacher then narrates: I am crossing the number of the charge, not the sign. The sign tells me who is positive and who is negative. The number tells me how many of each I need to balance the charges. After the second example teacher says: Now look at your prediction. Circle it — correct, partially correct, or needs revision. Tell your neighbour in one sentence what you got right and what you missed. [WAIT TIME: 90 seconds for paired exchange.]	The deliberate separation of observe-then-write prevents students from copying without understanding. The self-comparison of prediction against the observed method forces metacognitive reflection. The paired exchange immediately after consolidates vocabulary (subscript, crossover, simplify) in peer language before the teacher formalises it.	Teacher asks for a show of hands: How many found their prediction matched exactly? How many needed to revise? Teacher uses the ratio of revised hands to decide whether to add a second slow worked example (MgO) or proceed directly to the Explain phase practice set.

Explain Phase 📹 **VIDEO:** Second partial derivative test Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Teacher writes the four-step crossover method as a permanent classroom anchor on the board: Step 1 — write symbols and charges. Step 2 — cross the numbers. Step 3 — simplify. Step 4 — check charges balance. Students now copy this into their notebooks and then work through the following practice set in pairs, in order of increasing complexity. Set A (binary compounds, fixed valency): NaCl, MgO, $CaCl_2$, Al_2O_3. Set B (compounds with radicals): $FeSO_4$, $Fe_2(SO_4)_3$, $Ca(NO_3)_2$, NH_4Cl. For each compound the teacher has pre-written the Kenyan context on the board: NaCl — Malindi salt flats; MgO — antacid tablets sold in Nairobi pharmacies; $CaCl_2$ — Kericho tea-farm soil; Al_2O_3 — aluminium cookware common in Kenyan households; $FeSO_4$ — iron supplement tablets; $Fe_2(SO_4)_3$ — rust treatment product; $Ca(NO_3)_2$ — calcium nitrate fertiliser used on Kericho tea farms; NH_4Cl — dry-cell batteries sold in Nairobi markets. After 10 minutes of pair work, teacher projects the full answer key. Students self-mark using a red pen, writing a tick or a cross for each formula. For every cross, students write one sentence explaining the specific error (for example: I forgot to put brackets around SO_4 before adding the subscript 3).	Before pair work begins teacher says: For Set B you must use brackets whenever the radical needs a subscript greater than 1. If you are unsure, look at your valency table — the radical charge is in the table you built yesterday. [WAIT TIME: 10 minutes of uninterrupted pair work. Teacher circulates and uses targeted questions rather than giving answers: What is the charge on the sulfate radical? What does the crossover of those two numbers give you? Does 2 and 2 simplify?] During self-marking teacher says: Be honest with your red pen — a wrong formula that you mark correct will confuse you in the exam. Write the error sentence immediately while you still remember what you were thinking. [WAIT TIME: 2 minutes for error-sentence writing.]	The sequencing from binary to radical compounds provides scaffolded cognitive load. The Kenyan contexts anchor each abstract formula to a tangible object students have seen or used, reinforcing the anchoring phenomenon link. The error-pattern sentence transforms self-marking from a passive activity into metacognitive writing, which research shows improves retention of procedural knowledge.	Teacher collects a quick data point: How many students have zero errors in Set A? How many have errors only in Set B? How many have errors in both sets? Teacher uses this to form two groups for the next 3 minutes: students with Set A errors stay at the board with the teacher for a rapid re-teach of the crossover direction; students confident in Set A attempt an extension question: derive the formula for iron(III) hydroxide, $Fe(OH)_3$, using the OH^- radical.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Second partial derivative test	Students return to the Driving Question Board. They read the question posted in Lesson 6: How do we know which oxidation number to use? Students write a new sticky note responding to this question using evidence from	Teacher says: The question from Lesson 6 asked how we know which oxidation number to use. You now have a method that answers that. Write your sticky note as if you are answering a classmate who missed todays	Returning physically to the element cards from Lesson 1 closes a visible loop in the storyline — students see tangibly that their understanding has grown from sorting cards by appearance to being able to predict combining	Teacher reads the sticky-note responses on the DQB and checks whether students are linking oxidation number to valency correctly. Vague responses (for example: we use the number from the table) are gently challenged

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	todays lesson — they must reference at least one formula they derived and one Kenyan compound. A second new question is generated by the teacher prompt: Now that we can write formulae, what do we do when two of these compounds react with each other? Students post this new question on the DQB under the label Lesson 8 will answer this. A third prompt asks students to connect back to the original Mystery Element Card Challenge: If someone showed you Card 20 (calcium) and Card 17 (chlorine) today, what would you write on a new version of those cards that was not there before? Students write formula: $CaCl_2$ on a small card and attach it to the relevant element cards on the classroom wall display.	lesson — use the formula you are most proud of as your example. [WAIT TIME: 2 minutes individual writing.] Teacher then gestures to the element card wall display and says: Go back to the very first cards you sorted on Day 1. What new information can you add to those cards right now, with what you know today? [WAIT TIME: 90 seconds for students to physically add their formula card to the display.]	ratios from invisible charge patterns. The new DQB question about reactions creates forward momentum toward Lesson 8.	verbally: Which table? What does that number represent? This probes depth of understanding before students leave.
Model Building Phase 📹 **VIDEO:** Second partial derivative test Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Observation: Penny Oxidation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students turn to the element-card diagram they have been developing across the unit. On the back of their personal element card (one element assigned per student in Lesson 3), they add a new layer to the model: they draw the crossover diagram showing their element combining with a partner element chosen from the class set, write the resulting formula, and add a Kenyan context sentence. Students whose element is in Group 1 or 2 combine with a halogen partner; students whose element is a halogen combine with an alkali metal or alkaline earth metal partner; students with transition elements choose either O^{2-} or SO_4^{2-} as their partner. The updated model now shows: atomic number and mass (Lesson 1) — electron arrangement diagram	Teacher says: Your element card has been growing lesson by lesson. Today it gets its newest layer — the formula. This is the first time your element card shows something a chemist would actually write in a lab report or on a fertiliser bag. Make the crossover diagram clear enough that someone who missed todays lesson could follow your steps. [WAIT TIME: 4 minutes for model updating and verbal sharing.] Teacher circulates and listens for correct use of vocabulary: subscript, bracket, crossover, simplify, radical. Teacher offers a stretch prompt to early finishers: If your element formed a compound with the sulfate radical instead, what would the formula be? Show the crossover.	The cumulative element-card model makes the unit storyline tangible and personal — each student owns a physical record of how their understanding has built layer by layer from the anchoring phenomenon to formula derivation. The verbal explanation to a neighbour activates retrieval practice in a low-stakes peer setting.	Teacher selects three or four student model cards to place under the document camera briefly at the end of the phase, asking the class: Is the crossover diagram correct here? Are the brackets in the right place? Is the formula simplified? This whole-class gallery-style check normalises error discussion and gives the teacher live data on class-wide formula-writing accuracy to inform the Lesson 8 starting point.

	(Lesson 3) — ion charge and dot-cross diagram (Lesson 5) — valency and oxidation number (Lesson 6) — crossover diagram and formula (Lesson 7). Students share their updated model card with a neighbour and explain the crossover step verbally.			

D. TEACHER REFLECTION

After the lesson, ask yourself the following: Did students distinguish between crossing the number and crossing the sign — was the most common error a sign error or a subscript-direction error? Did the Kenyan contexts (Kericho tea farms, Nairobi dry-cell batteries, Malindi salt) feel genuine to students or superficial — could students name the context without being prompted? Did the self-marking and error-sentence writing reveal systematic misconceptions (for example, always forgetting brackets for radicals with subscript greater than 1) that need a brief recap at the start of Lesson 8? Were the two ability groups during the Explain phase re-teach moment well-matched — did the extension group genuinely stretch or did they wait? Review the DQB sticky notes tonight: do students link oxidation number to valency fluently, or are they treating the crossover as a mechanical trick without conceptual grounding? If the latter, plan a 5-minute conceptual re-anchor at the start of Lesson 8 before introducing word equations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the crossover method applied to calcium chloride and sodium chloride on the board, and observed how crossing the numerical values of ion charges generates the correct subscript ratio in the formula.
What did I learn?	Students learned that the crossover method uses valency (the magnitude of the ion charge) to determine the subscript of each ion in an ionic formula, that brackets must enclose radicals when the subscript is greater than 1, and that the resulting formula can be verified by checking that total positive charge equals total negative charge.
How does this explain the phenomenon?	The pattern is explained by the requirement for ionic compounds to be electrically neutral — the number of each ion type in the formula must be exactly the ratio that makes the sum of positive charges equal the sum of negative charges, which is precisely what the crossover method calculates from the valencies derived from periodic table position.

LESSON 8 (40 minutes): Writing and Balancing Chemical Equations	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will write and balance chemical equations for simple reactions drawn from Kenyan industrial and everyday contexts, connecting the requirement for balancing to the Law of Conservation of Mass.
Knowledge	Students know that a chemical equation shows reactants on the left and products on the right; that balancing coefficients (never changing subscripts) ensures equal numbers of each atom on both sides; and that this reflects the fact that atoms are neither created nor destroyed during a chemical reaction.
Skills	Students can convert a word equation to a symbol equation, use an atom inventory table to identify imbalances, and adjust coefficients systematically to produce a balanced equation for at least six reactions.
Attitudes	Students appreciate that precise symbolic representation is a universal scientific language that allows Kenyan chemists and farmers to communicate formulations accurately with the wider scientific community.
Key Inquiry Question	How do chemical bonds influence the properties of substances? — Balanced equations make bond-breaking and bond-forming visible at the symbolic level, showing exactly which substances are consumed and which are produced when bonds change.
Purpose in Storyline	In Lessons 5–7 students built ion diagrams, assigned oxidation numbers, and derived ionic formulae using the crossover method. Now they assemble those formulae into complete chemical equations, adding the conservation constraint that ties every formula to real atoms. This prepares the class to write the potassium-with-water equation needed for the Lesson 9 farmer scenario and to close the storyline chain: periodic table position → electron arrangement → ion → formula → balanced equation.
Safety Notes	No hazardous materials are handled in this lesson. All reactions are treated symbolically. If a teacher demonstration of magnesium burning is included as a motivating observation, standard precautions apply: do not look directly at the flame, keep combustibles away, use tongs and a heat-proof mat. In this lesson plan the reactions are addressed only on paper.

B. LESSON OVERVIEW

The lesson opens with a Predict phase in which students recall the formulae they derived in Lesson 7 and attempt to write a word equation for the burning of charcoal (carbon) — a phenomenon every Kenyan student recognises from nyama choma grills — before translating it to symbols. In the Observe phase the teacher models the three-step process (word equation → symbol equation → atom inventory → balanced equation) using two vivid Kenyan examples: sodium reacting with water (echoing the sodium card from the original Mystery Element Card Challenge) and the burning of limestone to make lime at the Athi River Bamburi Cement plant ($CaCO_3$ decomposition). In the Explain phase students receive a structured worksheet and independently practise balancing six equations, then peer-review using a colour-coded atom inventory table. The DQB is updated with the new question linking balancing to the Law of Conservation of Mass. In the Model Building phase students annotate one balanced equation with particle-level diagrams, connecting back to the element cards from Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students open notebooks to their	Display the nyama choma	Eliciting prior knowledge through a	Teacher scans mini-whiteboards

📹 **VIDEO:** [It's not only about the American Revolution, Leutze's Washington Crossing the Delaware] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Use the Percent Equation to Find the Base, b (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Lesson 7 valency table. The teacher projects a photograph of a nyama choma charcoal grill at a Nairobi roadside and asks: 'You can see carbon burning in air. In your notebook write (a) a word equation for this reaction and (b) try to turn it into a symbol equation using the formulae you already know.' Students work silently for 3 minutes. They then share predictions with a shoulder partner. Common predictions include: carbon + oxygen → carbon dioxide, with symbol attempts ranging from C + O → CO2 to C + O2 → CO2. Some students may write C2 for carbon, revealing the misconception that diatomic notation applies to all elements. Teacher asks pairs to hold up mini-whiteboards or open notebooks showing their symbol attempt.	photograph and say: 'Every time your family makes nyama choma, a chemical reaction is happening right in front of you. Before I show you anything, I want you to predict — in words first, then in symbols — what is happening chemically.' Allow 3 minutes of silent individual thinking, then 1 minute of pair discussion. Circulate and note three types of responses: (1) correct word equation with correct symbols, (2) correct word equation but incorrect symbols such as C2 or O instead of O2, (3) incomplete attempts. After sharing, say: 'I can see several different symbol equations on your boards. Today we are going to develop a reliable method so that any chemist in Kenya or anywhere in the world would write exactly the same equation.' Do NOT correct errors yet — let them stand as predictions to revisit in the Observe phase. WAIT TIME: after posing the prediction task, wait a full 3 minutes before accepting any responses. After asking pairs to share, wait 10 seconds before commenting.	familiar Kenyan phenomenon (charcoal burning) activates Lesson 7 formula knowledge and surfaces the key misconception that elemental solids such as carbon or sodium might need a subscript. Recording predictions in notebooks creates accountability and a baseline for self-correction later.	or open notebooks to tally: How many students wrote O2 correctly? How many used C2? How many left the product formula blank? This informs the emphasis needed in the Observe phase — specifically the distinction between diatomic molecules and monatomic elemental solids.
Observe Phase 📹 **VIDEO:** [It's not only about the American Revolution, Leutze's Washington Crossing the Delaware] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Teacher writes the three-step framework on the board: Step 1 — Write the word equation. Step 2 — Replace each substance with its correct formula (symbol equation, unbalanced). Step 3 — Build an atom inventory table and adjust coefficients until both sides match. The teacher then models this with two examples. Example 1: Sodium reacts with water to form sodium hydroxide and hydrogen gas (linking directly to the sodium card from the Mystery Element Card Challenge). The teacher writes the word equation, then the	For Example 1, write only the word equation first and ask: 'What formula did we derive in Lesson 7 for sodium hydroxide?' Pause for 10 seconds. Accept student responses before writing NaOH. Then write the unbalanced equation Na + H2O → NaOH + H2 and say: 'Let us count atoms on each side. I will make a table — help me fill it in.' Draw a four-column table: Element | Left side | Right side | Balanced? Read each element aloud and ask for counts by show of hands. When the table reveals H is unbalanced (2 on left,	The teacher uses a think-aloud modelling strategy combined with student call-and-response to make the invisible atom-counting process visible. Connecting Example 1 to the sodium Mystery Card and Example 2 to a landmark on the Nairobi–Mombasa highway embeds the abstract procedure in the anchoring phenomenon and in Kenyan geography, maintaining storyline coherence.	During call-and-response atom counting, the teacher notes which students give incorrect counts and which students correctly identify the unbalanced element before the table is complete. If fewer than half the class correctly counts hydrogen in Example 1, the teacher pauses for a second guided example before releasing students to independent practice.

Use the Percent Equation to Find the Base, b (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	unbalanced symbol equation 2Na + 2H2O → 2NaOH + H2, building the atom inventory table column by column on the board with student call-and-response for atom counts. Example 2: Calcium carbonate decomposes on heating to form calcium oxide and carbon dioxide — the exact process used at the Athi River Bamburi Cement works visible to students travelling on the Nairobi–Mombasa highway. CaCO3 → CaO + CO2 is already balanced; teacher uses this to show that sometimes no adjustment is needed and the inventory table confirms it. Students copy the atom inventory tables into their notebooks and annotate each step with marginal notes in their own words.	3 on right in the unbalanced version), say: 'We have a problem. Where did that extra hydrogen atom come from? Remember — atoms cannot be created. So what must we do?' WAIT TIME: 15 seconds. Guide students to adjust the coefficient of H2O to 2, then recount. For Example 2, say: 'This reaction happens every day at the Bamburi plant near Athi River — limestone in, lime out, and the CO2 goes into the atmosphere. Let us check if it is already balanced.' After building the inventory table and confirming balance, say: 'Not every equation needs adjustment — the table tells you for certain.' Critical teacher move: explicitly state 'We NEVER change a subscript to balance an equation. Changing a subscript changes the substance itself — NaOH becomes something entirely different if you write NaOH2. We only change the big number in front — the coefficient.' Write this rule in a box on the board.		
Explain Phase 📹 **VIDEO:** It's not only about the American Revolution, Leutze's Washington Crossing the Delaware Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Use the Percent Equation to Find the Base, b (Flexbook)	Students receive a structured worksheet containing six reactions to balance, all with Kenyan contexts: (1) Magnesium burning in air — magnesium ribbon is used in school labs across Kenya; (2) Calcium carbonate reacting with dilute hydrochloric acid — the effervescent test students will meet in qualitative analysis; (3) Iron reacting with oxygen to form iron(III) oxide — rust on Kisumu Port railings introduced in Lesson 6; (4) Potassium reacting with water to form potassium hydroxide and hydrogen — preparing for the Lesson 9 Nakuru farmer scenario; (5) Sodium carbonate reacting with hydrochloric acid — washing soda	Distribute the worksheet and say: 'You have 12 minutes. Work independently — no partner discussion yet. Use the three-step method exactly as we practised. If you are stuck, go back to your atom inventory table and recount before changing anything.' Circulate and use targeted prompts rather than giving answers: 'How many oxygen atoms do you have on the right? Count again.' and 'Have you changed a subscript or a coefficient? Point to the number you changed.' After the peer-marking, ask: 'Show me by thumbs up or down — did your partner change any subscripts?' If many	The structured worksheet with printed atom inventory tables scaffolds the procedure while the Kenyan contexts (Kisumu rust, washing soda, Nakuru potassium) maintain relevance. Colour-coded peer marking makes error patterns concrete and non-threatening, transforming marking into a learning activity rather than a judgment. The debrief on the two most common errors consolidates the key conceptual distinction between coefficient and subscript.	Peer-marked worksheets are collected at the end of the lesson. The teacher scans for: proportion of students with all six balanced correctly (target: 70 percent); frequency of the subscript-change error; frequency of omitting brackets around radicals such as OH in Ca(OH)2. Results inform the opening review in Lesson 9.

Source: CK-12 🔍 Search ARES for similar readings	used in Kenyan households; (6) Hydrogen gas reacting with oxygen to form water — included to balance a reaction with a diatomic product. For each reaction students first write the word equation, then the symbol equation, then complete a printed atom inventory table, then state the balanced equation and the coefficient used. After 12 minutes of independent work, students exchange worksheets with a different partner (not the shoulder partner used in Predict) and mark using a colour-coded key: green for correct, amber for coefficient error, red for subscript changed illegally. Pairs discuss any red or amber markings before the teacher leads a brief whole-class debrief on the two most common errors seen.	thumbs point down (errors found), project the most common error anonymously and ask the class to diagnose it: 'What rule did this student break? How do we fix it without changing the subscript?' WAIT TIME: after asking for diagnosis of the anonymous error, wait 20 seconds for thinking before accepting responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** It's not only about the American Revolution, Leutze's Washington Crossing the Delaware Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Use the Percent Equation to Find the Base, b (Flexbook) Source: CK-12 🔍 Search ARES for similar	With 5 minutes remaining, the teacher draws attention to the DQB at the front of the room. Students revisit the sticky notes posted in earlier lessons, particularly the Lesson 4 note asking 'Why does reactivity increase down Group 1?' and the Lesson 7 note asking 'How do we know which oxidation number to use?' The teacher asks: 'Balancing an equation means the number of atoms on both sides must be equal. Which law in science does that remind you of?' Students think for 30 seconds, then one student from each group posts a new sticky note. The intended new question is: 'Why must equations be balanced — what law does this connect to?' If students do not generate this question independently, the teacher models posting it and labels it as a	Point to the DQB and say: 'Every lesson we have added new questions. Today we discovered that we must balance equations. Why? What deep principle forces us to do this?' Allow 30 seconds of silent thinking. Then say: 'Write your question on a sticky note and post it. If your question is the same as a classmate post it on top — that shows it is an important shared question.' After posting, read the new questions aloud and say: 'The question about conservation of mass will be answered in your physics strand as well — this is one of those ideas that cuts across all of science. Next lesson we will use these balanced equations to solve a real problem for a Kenyan farmer.' WAIT TIME: 30 seconds of silence after posing the conservation-of-mass prompt, with	Returning to the DQB at the end of each lesson reinforces the idea that science proceeds by asking questions and accumulating evidence. Linking the new question about conservation of mass to the driving question ('Why do elements behave so differently?') shows students that even the symbolic representation of reactions is constrained by the same underlying atomic structure they have been studying throughout the unit.	The quality and specificity of student-generated DQB questions reveals depth of understanding. Vague questions such as 'Why do we balance?' indicate surface engagement; specific questions such as 'Does the mass of the products always equal the mass of the reactants in the sodium-water reaction?' indicate conceptual engagement with conservation. Teacher notes the split for differentiation in Lesson 9.

readings	question to investigate. A second prompt: 'Looking at the sodium-plus-water equation we balanced today, can you see which bonds were broken and which new bonds formed? Post a question if something is puzzling.' This bridges to Lesson 9 and Sub-Strand 1.4.	no further teacher talk, so students can genuinely formulate their own questions.		
Model Building Phase 📹 **VIDEO:** It's not only about the American Revolution, Leutze's Washington Crossing the Delaware Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Use the Percent Equation to Find the Base, b (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 3 minutes students open their notebooks to their element card diagram for sodium (drawn or pasted in Lesson 3) and add one new annotation layer: the balanced equation for sodium reacting with water, written underneath the electron shell diagram. They draw an arrow connecting the Na atom diagram to the Na in the balanced equation and label it 'loses 1 electron → Na^+'. This mini-model — atomic structure connected to balanced equation — is the seed of the full annotated poster students will produce in Lesson 10. Students who finish early are asked to add a second annotation for calcium carbonate decomposition, connecting the Ca^{2+} ion diagram from Lesson 5 to the CaO in the balanced equation.	Say: 'In Lesson 1 you sorted mystery cards. In Lesson 3 you drew electron shells. In Lesson 5 you drew ions. In Lesson 7 you wrote formulae. Today you wrote balanced equations. Now connect them: open your sodium card diagram and write the balanced equation underneath it. Draw a line from the Na atom to the Na in the equation.' Circulate and prompt: 'Your sodium loses one electron. Where does that electron go in sodium hydroxide? Can you show that?' This question is intentionally open-ended and will be answered fully in Sub-Strand 1.4; the goal here is to create productive curiosity. WAIT TIME: after posing the extension question about where the electron goes, wait 15 seconds and then say: 'That is a great question to leave open — we will answer it fully when we study chemical bonding.'	The mini-model annotation closes the lesson by explicitly linking today's learning to the cumulative storyline. Making the connection visible in the notebook — as a physical annotated diagram — prepares students for the Lesson 10 gallery-walk poster and prevents the lesson from feeling like an isolated skill exercise.	Teacher circulates and checks that at least the balanced equation and the electron-loss annotation are present on the sodium diagram. Students who have not reached this point by the end of class are identified for a brief catch-up check at the start of Lesson 9.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did the majority of students successfully distinguish between changing a coefficient and changing a subscript — evidenced by the peer-marked worksheets? (2) Were the Kenyan contexts (nyama choma charcoal, Athi River lime, Kisumu rust, Nakuru potassium) enough to keep students engaged or did some contexts need more visual support? (3) Did the DQB question about conservation of mass emerge from students themselves or did the teacher need to supply it — if supplied, plan a brief revisit at the start of Lesson 9. (4) Were higher-achieving students sufficiently stretched by the extension annotation task, or should Lesson 9 open with a more challenging equation such as the combustion of ethanol used in Kenyan biogas stoves?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that the atom inventory table reveals imbalances in symbol equations for real reactions including sodium reacting with water and calcium carbonate decomposing at the Athi River cement plant.
What did I learn?	Students learned the three-step method for writing and balancing chemical equations: (1) write the word equation, (2) replace substances with correct formulae to give the unbalanced symbol equation, (3) adjust coefficients — never subscripts — using an atom inventory table until atoms are equal on both sides.
How does this explain the phenomenon?	The requirement to balance equations is explained by the principle that atoms are conserved in chemical reactions — no atom is created or destroyed, only rearranged — which means the total number of each type of atom must be identical in the reactants and the products, connecting to the Law of Conservation of Mass.

LESSON 9 (40 minutes): Consolidation and Extended Problem Solving: From Periodic Table Position to Balanced Equation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students apply the full conceptual chain — periodic table position to electron arrangement to ion formation to chemical formula to balanced equation — using a real Kenyan agricultural scenario to consolidate learning from Lessons 1 through 8.
Knowledge	Students will be able to: (1) identify the chemical families and periodic table groups of all elements present in urea (CH4N2O), calcium ammonium nitrate (Ca(NO3)2·NH4NO3), and muriate of potash (KCl); (2) derive the formulae of ionic compounds from valencies using the crossover method; (3) write and balance the equation for the reaction of potassium with water.
Skills	Analysis of compound composition by periodic table position; derivation of chemical formulae from valency tables; writing and balancing chemical equations using an atom inventory table; collaborative problem solving; critical peer evaluation of balanced equations.
Attitudes	Curiosity and persistence in tackling multi-step chemical problems; appreciation of the practical value of chemistry in Kenyan farming; respect for peers contributions during group critique; confidence in using models built over the unit.
Key Inquiry Question	How do chemical bonds influence the properties of substances? — Students see how the bonding implied by valency and ion formation directly determines the formula of a compound and the stoichiometry of its reactions.
Purpose in Storyline	This lesson is the integration checkpoint of the evidence-gathering sequence. Students revisit the Mystery Element Card Challenge and demonstrate that what began as a puzzling sorting exercise can now be decoded completely: an element card reveals periodic table position, which reveals electron arrangement, which predicts ion formation and valency, which gives the formula, which can be used in a balanced equation. The scenario card about a Nakuru farmer ties all prior lessons into a single coherent narrative before the gallery walk and DQB closure in Lesson 10.
Safety Notes	No hazardous chemicals or live demonstrations in this lesson. All work is pencil-and-paper and collaborative discussion. Remind students that potassium metal reacts violently with water in real life — the balanced equation is a theoretical exercise only and must never be attempted without specialist laboratory conditions.

B. LESSON OVERVIEW
Lesson 9 is a 40-minute collaborative consolidation lesson structured around a scenario card describing a Nakuru farmer choosing three fertilisers. Students move through a Predict-Observe-Explain cycle in which they first predict the chemical families and likely reactions before working through the full analytical chain as a group. The lesson surfaces and corrects residual misconceptions from Lessons 3 through 8, advances the Driving Question Board with synthesised evidence, and has students build a chain model linking all five conceptual steps. The Kenyan agricultural context (fertiliser use in the Rift Valley) grounds abstract chemistry in a situation students recognise from geography and everyday life.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students receive the Nakuru Farmer Scenario Card (one per	Teacher distributes scenario cards and says: 'Before we discuss	Individual prediction before group discussion prevents anchoring	Teacher reads notebooks during silent circulation and mentally

📹 **VIDEO:** [Valence Electrons and the Periodic Table - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Writing the Periodic Table (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	group of four). The card reads: 'Kamau, a maize farmer in Nakuru, wants to apply three fertilisers this season: urea (CH4N2O), calcium ammonium nitrate (Ca(NO3)2·NH4NO3), and muriate of potash (KCl). He also notices that the potassium bag has a warning: keep away from water. Help Kamau understand the chemistry of his fertilisers.' Before any group discussion, each student individually writes in their notebook: (a) Which chemical family does each element in these three fertilisers belong to? (b) Which group in the periodic table supplies potassium, and what do you expect will happen when it meets water? (c) What formula would you write for potassium chloride if you had to derive it from valency? Students take 4 minutes to write individual predictions. The teacher circulates silently, reading notebooks without comment. After 4 minutes, students share predictions within their group and note agreements and disagreements on a mini-whiteboard or piece of paper.	anything as a class, I want each of you to commit to your own prediction in your notebook. Use everything you have built since Lesson 1 — your element cards, your valency table, your equation-balancing steps. You have 4 minutes working alone.' [WAIT TIME: 4 minutes of silent individual writing — teacher circulates and reads notebooks without speaking or nodding, noting common errors for later targeted questioning.] After 4 minutes: 'Now take 2 minutes in your group — compare predictions. Where do you agree? Where do you disagree? Write the disagreements on your group paper — those are the most interesting places.' [WAIT TIME: 2 minutes of group comparison.] Teacher then asks one group: 'Tell me one disagreement your group had about which family carbon belongs to.' Listens to response without evaluating, then asks the class: 'Does another group have a different view? Raise your hand if your group placed carbon in a different family.' This surfaces the common misconception that carbon, being in Period 2, is a metal. Teacher does NOT resolve the disagreement yet — says: 'Hold that question. The Observe phase will give you the evidence to decide.'	bias — weaker students cannot simply copy the loudest voice. Recording disagreements explicitly prepares students for productive conflict resolution during the Observe phase. Connecting the prediction task to the original element cards (Lesson 1) reminds students that the same 20 cards they struggled to sort in Lesson 1 now make complete sense.	tallies: (1) Can students correctly place K in Group 1 (alkali metals)? (2) Do students correctly assign valency of 1 to K and 1 to Cl, giving KCl? (3) Do students recall that Group 1 metals react with water to give a metal hydroxide and hydrogen? Common errors noted: placing N in a metal group; confusing the NH4+ ion as a metal; writing K2Cl instead of KCl. These errors are targeted in the Explain phase.
Observe Phase 📹 **VIDEO:** [Valence Electrons and the Periodic Table - Overview] Source: CK-12 🔍 Search ARES for similar videos	Groups receive a Resource Envelope containing: (A) a mini periodic table with the first 20 elements and their groups highlighted; (B) the valency and oxidation number table from Lesson 6; (C) the atom inventory table template from Lesson 8; (D) their own element cards from Lesson 1 for H, C, N, O, K, Ca,	Teacher says: 'Your envelope has everything you need — the periodic table, valency table, atom inventory template, and your original element cards. These are the same tools that guided you through Lessons 3 to 8. Open the envelope and start Task 1 now.' [WAIT TIME: teacher allows 6 full minutes for Task 1 before	The Resource Envelope structures the observe phase so that students are reading real data (the periodic table, valency table) rather than receiving information from the teacher. Comparing potassium-plus-water to sodium-plus-water (Lesson 8) uses analogical reasoning — a key scientific practice — and reinforces	Teacher scans group analysis tables for: correct identification of C, N, O as non-metals in Groups 14, 15, 16 respectively; correct identification of K as Group 1 alkali metal and Ca as Group 2 alkaline earth metal; correct identification of Cl as Group 17 halogen. Teacher checks KCl derivation for correct crossover method. Teacher

📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	and Cl. Task 1 (6 minutes): Each group locates every element in the three fertilisers on the mini periodic table and completes a group analysis table with columns: Element | Atomic Number | Group | Period | Chemical Family | Outer Electrons | Typical Valency. Task 2 (5 minutes): Groups use the crossover method to derive the formula of KCl from first principles (K has valency 1, Cl has valency 1 — crossover gives K1Cl1, simplified to KCl) and then attempt to derive the formula of Ca(NO3)2, treating NO3 as a radical with valency 1 (charge 1-). Task 3 (5 minutes): Groups write the word equation for potassium reacting with water, then convert to a symbol equation, then use an atom inventory table to balance it. They compare their balanced equation to the one derived for sodium plus water in Lesson 8 and note similarities and differences.	speaking again. Circulates and observes without intervening unless a group is completely silent and stuck, in which case teacher points silently to the relevant row of the periodic table without giving the answer.] After Task 1: 'What do you notice about where carbon, nitrogen, and oxygen sit in the periodic table relative to the metals potassium and calcium?' [WAIT TIME: 20 seconds for students to look before a group responds.] After Task 2: Teacher asks one group to write their KCl derivation on the board. Teacher then asks: 'Did everyone get KCl? If you wrote K2Cl or KCl2, look at your valency table again — what valency does chlorine have when it forms a chloride?' [WAIT TIME: 15 seconds.] After Task 3: Teacher asks a group to read their balanced equation for potassium plus water. If any group writes K + H2O → KOH + H without balancing, teacher says: 'Count your hydrogen atoms on each side using the atom inventory table. What do you find?' [WAIT TIME: 20 seconds for students to recount.]	the group 1 family trend discussed in Lesson 4. Using the original Lesson 1 element cards as part of the envelope physically reconnects this lesson to the anchoring phenomenon.	checks atom inventory tables for the equation 2K + 2H2O → 2KOH + H2, looking specifically for correct coefficient of 2 on both K and H2O and the molecular formula H2 (not 2H) for hydrogen gas.
Explain Phase 📹 **VIDEO:** Valence Electrons and the Periodic Table - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES	Each group prepares a 2-minute group presentation covering: (1) the completed group analysis table for all elements in the three fertilisers; (2) the derivation of KCl and Ca(NO3)2 from valencies; (3) the balanced equation 2K + 2H2O → 2KOH + H2 with an annotated atom inventory table. While one group presents, all other groups are peer evaluators: they use a Peer Critique Card with three sentence starters — 'The formula is correct because...', 'The equation is balanced because...',	Teacher says: 'Group 2, you are up first. The rest of us are peer evaluators. Use your Peer Critique Card. You have 2 minutes, Group 2.' [WAIT TIME: full 2 minutes of uninterrupted presentation.] After presentation: 'Peer evaluators — starting with the first sentence starter: who has a comment about whether the formula is correct?' [WAIT TIME: 10 seconds for hands.] Selects one student, listens fully, then asks presenting group: 'Do you agree with that critique? Why or why not?' [WAIT	Peer critique using structured sentence starters elevates the quality of feedback beyond 'it is right' or 'it is wrong' by requiring students to cite evidence. The five-step chain written on the board serves as a visible consolidation model that students can reference in Lesson 10. The NH4+ misconception is addressed through a deliberate Socratic question rather than direct correction, preserving student agency.	Teacher listens to group presentations for: correct chemical family identification; correct use of crossover method; correct balanced equation with H2 (not H) for hydrogen gas. Teacher reads Peer Critique Cards collected at the end of the phase to check whether students can articulate why an equation is balanced (conservation of atoms) rather than just stating that it is. This informs the final consolidation needed in Lesson 10.

[for similar readings]()	and 'I suggest a revision because...'. After each presentation (approximately 2 minutes per group — two groups present in the time available), the class discusses the most interesting critique raised. Teacher facilitates by asking targeted questions based on errors noted during the Observe phase. Teacher explicitly addresses the misconception that NH4+ is a metal (it is the ammonium radical, a polyatomic ion, not an element). Teacher also draws explicit attention to the parallel between K and Na reactions with water, asking: 'What does this similarity tell you about the power of knowing an element is in Group 1?' Teacher writes the chain on the board: Group 1 position → 1 outer electron → loses 1 electron → valency 1 → forms K+ ion → KCl formula → 2K + 2H2O → 2KOH + H2. Students copy this chain into their notebooks as the core model for the lesson.	TIME: 15 seconds.] After both presentations and critiques, teacher says: 'I want to address something I noticed in several notebooks. Some groups wrote NH4+ as if ammonium were a metal element. Look at your periodic table — is ammonium listed there?' [WAIT TIME: 10 seconds.] 'Ammonium is a radical — a group of atoms that travel together and carry a charge. It is not an element and has no place on the periodic table. It behaves like a Group 1 ion because it carries a 1+ charge, but it is NOT potassium.' Teacher then writes the five-step chain on the board, reading each step aloud, pausing after each for students to write. Teacher concludes: 'This chain is the answer to our Driving Question. From position — to electrons — to ions — to formula — to equation. Kamau the farmer may not know this chain, but every formula on his fertiliser bag is built on it.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [DNA as the "transforming principle"]() Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos]() 📖 **HTML:** [Writing the Periodic Table (PLIX)]() Source: CK-12 [🔍 Search ARES]()	Students return to the class Driving Question Board (posted since Lesson 1). In pairs, they read through the sticky notes accumulated over Lessons 1 to 8. Each pair selects one question from the board and writes a brief answer (2 to 3 sentences) in their notebook using evidence from today's lesson. Suggested pairings offered by the teacher: pairs who focused on the Nakuru scenario answer 'Why does reactivity increase down Group 1?'; pairs who found the NH4+ issue interesting answer 'How do we know which oxidation number to use?'; pairs who wrote the balanced equation answer 'Why	Teacher says: 'Walk to the DQB. Find the question your group or another group posted that today s lesson can now answer. Write a 2-sentence answer in your notebook — use evidence from the scenario card, the periodic table, or the balanced equation.' [WAIT TIME: 3 minutes of quiet reading and writing at the board.] After 3 minutes: 'Pair up with someone next to you and read your answers to each other. Does your partner s evidence make your answer stronger?' [WAIT TIME: 1 minute.] Then: 'Who wants to share an answer that uses the five-step chain we wrote on the board?' Selects two pairs. After sharing:	Returning to the DQB makes the intellectual journey of the unit visible and satisfying. Students experience epistemic closure on questions they themselves posted, which builds metacognitive awareness of how scientific understanding develops. The new bridging question maintains productive curiosity and prevents the sense that chemistry is a closed, completed subject.	Teacher reads the notebook answers written at the DQB to check whether students can connect today s integrated task back to specific earlier lessons. Look for: use of the word 'conservation' in answers about balanced equations; reference to 'outer electrons' or 'Group 1' in answers about reactivity trends; use of the term 'radical' or 'polyatomic ion' in answers about oxidation numbers.

for similar readings	must equations be balanced — what law does this connect to?' Two pairs share their answers aloud. The teacher then adds one new sticky note to the DQB in a distinctive colour: 'From one element card, we can now trace all the way to a balanced equation — what else could we predict if we knew only an element s position?' This bridges to Lesson 10 and to Sub-Strand 1.4 on Chemical Bonding.	teacher adds the new bridging sticky note and reads it aloud. 'This question will be the starting point for Lesson 10 and for what comes next in Sub-Strand 1.4. Keep it in mind.'		
Model Building Phase 📹 **VIDEO:** Valence Electrons and the Periodic Table - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 6 minutes, each student individually draws the five-step chain model in their notebook as a annotated flow diagram: Box 1 — Element Card (atomic number, Kenyan context); Arrow — leads to: Box 2 — Periodic Table Position (group, period, family); Arrow — leads to: Box 3 — Electron Arrangement (outer electrons, shell diagram); Arrow — leads to: Box 4 — Ion Formation and Valency (cation or anion, charge, crossover formula); Arrow — leads to: Box 5 — Balanced Chemical Equation (with atom inventory table). Students use potassium as their worked example, filling in each box with the specific information from today s lesson: K, Group 1, alkali metal, 2-8-1 arrangement, K^+ ion, valency 1, KCl formula, $2K + 2H_2O \rightarrow 2KOH + H_2$. Students annotate the chain with a brief note explaining why each arrow is justified — for example, the arrow from Box 2 to Box 3 is annotated: 'Group number tells us number of outer electrons.' Students who finish early add a second example using calcium, tracing through to $Ca(NO_3)_2$. The teacher collects the notebooks at the end of the	Teacher says: 'In your notebook, draw five boxes connected by arrows. This is your personal model — the one you will bring to the gallery walk in Lesson 10. Label each box as I show you on the board.' [Teacher writes the five box labels on the board.] 'Fill in each box using potassium as your example. Use the specific information from today — not a generic description, but the actual numbers and symbols for potassium.' [WAIT TIME: 5 minutes of individual silent work. Teacher circulates and reads diagrams, placing a small dot next to any arrow annotation that is missing or vague. Does not correct verbally — students will refine in Lesson 10.] After 5 minutes: 'At the bottom of your model, write one sentence that begins: I used to think sorting element cards was just about appearance, but now I know... Finish that sentence.' [WAIT TIME: 30 seconds.] Teacher collects notebooks.	The five-box chain model is the physical product that students will bring into Lesson 10 s gallery walk. Drawing it individually after group work ensures that every student has personal ownership of the integrated model. The sentence stem 'I used to think... but now I know...' is a reflective routine that makes the conceptual shift from Lesson 1 to Lesson 9 explicit and memorable, completing the narrative arc of the Mystery Element Card Challenge.	Teacher reviews collected notebooks before Lesson 10 and checks: (a) all five boxes are labelled correctly; (b) each arrow annotation gives a chemical reason (not just draws an arrow); (c) the balanced equation is written correctly with coefficient 2 on K, H_2O, and KOH, and H_2 for hydrogen gas; (d) the sentence stem reveals genuine conceptual shift from visual/physical sorting to electron-based understanding. Students whose notebooks show gaps in boxes 3 or 4 receive a written prompt: 'Revisit your electron arrangement table from Lesson 3 and your ion diagrams from Lesson 5 before tomorrow.'

	lesson and returns them before Lesson 10 with a tick, a target, and a bridging note.			

D. TEACHER REFLECTION

After the lesson, consider: (1) Did all groups correctly place C, N, and O as non-metals without teacher intervention, or did several groups still confuse their family? If the non-metal identification was a common problem, spend 3 minutes at the start of Lesson 10 revisiting the metalloid staircase on the periodic table. (2) Did the NH4+ misconception reappear in multiple groups despite being addressed in Lesson 6? If so, prepare a targeted warm-up for Lesson 10 asking students to list three differences between an element and a radical. (3) Were students able to write the Peer Critique sentences with chemical evidence, or did they default to vague statements like the formula looks right? If the latter, provide modelled examples of evidence-based critique at the start of the gallery walk in Lesson 10. (4) Did the five-step chain model feel rushed in the final 6 minutes? If so, consider assigning the calcium extension example as a brief homework task to be brought to the Lesson 10 gallery walk. (5) Review the DQB answers written in notebooks: which questions from Lessons 1 to 8 still remain genuinely unanswered? These should be highlighted during the DQB closure in Lesson 10 and explicitly bridged to Sub-Strand 1.4 on Chemical Bonding.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that urea, calcium ammonium nitrate, and muriate of potash each contain elements from identifiable chemical families on the periodic table (non-metals C, N, O; alkali metal K; alkaline earth metal Ca; halogen Cl). They also observed that the balanced equation for potassium reacting with water follows the same pattern as sodium reacting with water, consistent with both being in Group 1.
What did I learn?	Students learned that the complete chain from periodic table position to balanced equation can be applied to real Kenyan agricultural chemicals. They confirmed that valency derived from electron arrangement correctly predicts compound formulae, and that balancing equations requires conservation of atoms on both sides (consistent with the Law of Conservation of Mass from Lesson 8).
How does this explain the phenomenon?	Students explained that an element s position in the periodic table (its group) determines its number of outer electrons, which determines whether it forms a cation or anion and with what valency. That valency determines the formula of its ionic compounds via the crossover method, and those formulae are the building blocks of balanced chemical equations — all of which can be read from the element cards that seemed purely puzzling in Lesson 1.

LESSON 10 (80 minutes): Model Gallery Walk, DQB Closure & Unit Reflection	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate the entire unit storyline by producing a final annotated model poster, completing the Driving Question Board, and demonstrating competency through a summative prediction task.
Knowledge	Students know that an element's position in the periodic table is determined by its atomic number and electron arrangement; that electron arrangement drives ion formation, valency, and oxidation number; and that these properties allow prediction of compound formulae and balanced equations.
Skills	Students can trace the complete storyline from element card → periodic table position → electron arrangement → ion formation → compound formula → balanced equation; give and receive structured peer feedback; and apply unit knowledge to an unseen element.
Attitudes	Students appreciate how the invisible internal structure of atoms (electron arrangement) explains the visible patterns on the element cards from Lesson 1, and value the collaborative process of building scientific understanding over time.
Key Inquiry Question	How do chemical bonds influence the properties of substances?
Purpose in Storyline	This is the final lesson of Sub-Strand 1.3. Students synthesise all evidence gathered across Lessons 1–9 into one cohesive annotated model poster using sodium chloride from the Malindi salt flats as the case study. The Driving Question Board is formally closed, with open questions explicitly bridged to Sub-Strand 1.4 (Chemical Bonding). A summative written task assesses whether each learner can independently apply the complete reasoning chain to an unseen element — resolving the original Mystery Element Card Challenge.
Safety Notes	No hazardous materials are used in this lesson. Ensure gallery walk movement is orderly to prevent crowding or damage to posters. Sticky notes and markers should be shared respectfully.

B. LESSON OVERVIEW

Students open by revisiting their original Lesson 1 element card groupings and articulating how their thinking has changed. They then work in groups to produce a final annotated model poster tracing the full unit storyline using sodium chloride (from Malindi salt flats) as the anchor case study. A structured Gallery Walk follows, with peers leaving 'Glow and Grow' sticky-note feedback. The class reconvenes to formally close the Driving Question Board — celebrating answered questions and explicitly flagging open questions as the bridge to Sub-Strand 1.4 (Chemical Bonding). The lesson ends with an individual summative written task in which students receive an unknown element's electron arrangement and must predict its group, period, valency, ion type, and write the formula of its compound with chlorine — the ultimate resolution of the anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Common polyatomic ions]	Each student retrieves their Lesson 1 written prediction about how to group the 20 element cards. Working individually for 3	Distribute students' Lesson 1 prediction sheets (kept in teacher's folder). Say: 'You are going to meet your Lesson 1 self today.	Metacognitive Annotation — students physically mark up their own prior thinking in a contrasting colour. This makes conceptual	Observable indicator: Student annotations correctly name electron arrangement (not mass or appearance) as the key

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	minutes, they annotate their original prediction in a different colour pen, writing one sentence explaining what they now know that they did not know in Lesson 1. Pairs then share annotations for 2 minutes. The class hears 4–5 cold-called responses, identifying the biggest shift in their thinking. Teacher records key 'before → after' contrasts on the board (e.g., 'I grouped by mass → now I group by electron arrangement and periodic table period/group').	Read what you wrote then — do not judge it, just read it.' Allow 2 minutes of silent reading. Then say: 'In a different colour, write one sentence: What do you know NOW that your Lesson 1 self did not?' Allow 3 minutes. Cold-call 5 students using the class register (rows 1, 3, 5, 7, 9). Ask: 'What is the single biggest shift in your thinking from Lesson 1 to today?' WAIT TIME: 12 seconds after each student speaks before moving on. Record two-column 'Before → After' notes on the board visibly. Close by asking: 'How does knowing electron arrangement finally explain why those element cards were so puzzling in Lesson 1?' WAIT TIME: 15 seconds. Accept 2 responses.	change visible and personal, helping learners recognise the distance they have travelled and building confidence before the summative task.	explanatory factor. Teacher scans 5–6 sheets during cold-call responses. Flag any student who still references mass or colour groupings for targeted support during Phase 2 group work.
Observe Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 construct a large A2 poster tracing the complete unit storyline using sodium chloride ($NaCl$) from the Malindi salt flats as the case study. The poster must include six labelled sections arranged as a flow: (1) The Element Card — Na and Cl original card data (atomic number, mass number, Kenyan context photo sketch); (2) Periodic Table Position — period and group for each; (3) Electron Arrangement — shells and s/p notation for Na and Cl atoms; (4) Ion Formation — dot-and-cross diagrams showing Na^+ and Cl^-, with oxidation numbers and valencies labelled; (5) Formula Derivation — using valency crossover to arrive at $NaCl$; (6) Balanced Equation — e.g., $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$ with state symbols. A seventh optional section invites groups to sketch the Malindi salt flat context and annotate why NaCl forms	Distribute A2 paper, markers, and the poster scaffold template (six labelled boxes pre-printed). Say: 'This poster is the scientific story you have been building for ten lessons. Each box must connect to the next — an arrow between boxes must have a label explaining WHY we move from one step to the next.' Circulate every 4 minutes. At each group, ask one probing question from this list: (a) 'Why does Na lose one electron rather than gaining seven?' (b) 'How does the group number of Cl tell you its valency?' (c) 'What does the coefficient 2 in front of Na in the equation represent?' (d) 'If a Malindi salt-flat worker asked you why sea water leaves salt behind when it evaporates, what would you say?' WAIT TIME: 10 seconds after each question before accepting a response. At 15 minutes, give a 5-minute warning. At 18 minutes	Storyline Poster (Sequential Flow Modelling) — the six-section structure mirrors the epistemic progression of the unit, making the causal chain (electron arrangement → observable properties → compound formation) visually explicit. Labelled arrows between sections force students to articulate mechanisms, not just recall facts.	Observable indicators: (1) Arrow labels between sections use causal language ('because,' 'therefore,' 'this causes'); (2) Dot-and-cross diagrams correctly show 8 electrons in Na^+ outer shell and 8 in Cl^-; (3) Balanced equation has correct coefficients and state symbols. Teacher photographs any poster with errors for whole-class discussion in Phase 5.

	there. Groups divide tasks: one student per section, with a 'quality checker' role rotating every 5 minutes.	say: 'Finalise your arrows and labels — arrows are as important as the boxes.'		
Explain Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Writing the Periodic Table (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Posters are displayed around the classroom walls. One group member stays at their poster as a 'docent' to answer questions; the other three rotate clockwise, spending 3 minutes at each of three other groups' posters. Rotating students carry two colours of sticky notes: yellow ('Glow' — something scientifically strong or clearly explained) and pink ('Grow' — a specific suggestion or a question they cannot answer from the poster alone). Each rotating student must leave at least one yellow and one pink note per poster visited. After 12 minutes of rotation, all students return to their own poster, read their sticky notes silently for 2 minutes, and the docent briefs the group on the most common feedback received.	Before the walk, model a Glow and a Grow note on a projected sample poster: 'Here is a Glow: "Na electron arrangement [2,8,1] is correctly linked to Group I — well done." Here is a Grow: "The arrow between Ion Formation and Formula has no label — why does Na^+ and Cl^- give NaCl and not Na_2Cl?" Notice — a Grow is a question or a specific suggestion, never just "good" or "wrong."' During the walk, circulate and prompt quiet students: 'What is one thing on this poster that surprises you or that you would have drawn differently?' After rotation, cold-call 3 docents: 'What was the most common Grow note on your poster?' WAIT TIME: 10 seconds. Record the top 3 recurring scientific misconceptions on the board for Phase 4 discussion. Say: 'These Grow notes are now OUR class questions — let us address them together.'	Structured Peer Critique (Gallery Walk with Criterion-Referenced Feedback) — by separating 'Glow' from 'Grow,' students are trained to give evidence-based feedback rather than evaluative judgements. The docent role builds communication and the ability to defend scientific claims — a core NGSS Science and Engineering Practice (Obtaining, Evaluating, and Communicating Information).	Observable indicators: (1) Pink 'Grow' notes reference specific scientific content (not aesthetics); (2) Docents can explain their group's reasoning when questioned; (3) Teacher's board list of top misconceptions informs the DQB closure discussion in Phase 4. Flag any group whose Grow notes reveal persistent errors in ion formation or equation balancing.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Writing the Periodic Table]	The class gathers facing the Driving Question Board displayed since Lesson 1. The teacher reads each question card aloud. Students vote silently (thumbs up = answered, thumbs sideways = partially answered, thumbs down = still open) for each question. For each 'thumbs up,' the class nominates which lesson and which evidence answered it, and the card is moved to a 'ANSWERED' column. For 'thumbs sideways' cards, students suggest what further evidence would be needed.	Stand beside the DQB. Say: 'Ten lessons ago, you could not explain why sodium and chlorine — so different from each other — end up together in the salt on your ugali. Let us see how far we have come.' Read each DQB card clearly. After each vote, cold-call one student from the majority group: 'Which lesson gave us the evidence for this?' WAIT TIME: 12 seconds. For the driving question itself — 'Why do the 118 elements behave so differently, and how can knowing an element's position in the	DQB Consensus Mapping — publicly sorting question cards into 'Answered,' 'Partially Answered,' and 'Bridge' columns makes the epistemological structure of the unit visible. Students see that science progresses by answering one question and generating the next, mirroring authentic scientific practice (NGSS SEP: Asking Questions and Defining Problems).	Observable indicators: (1) Students correctly match evidence to answered questions; (2) Driving question summary sentences in notebooks include 'electron arrangement' and 'periodic table position' as key terms; (3) Bridge questions generated by students are conceptually coherent with Sub-Strand 1.4 content (bonding, attraction between ions). Teacher collects 5 notebooks at random to check summary sentences.

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	For 'thumbs down' / open questions — particularly those touching on WHY ions attract each other and WHY NaCl is solid at room temperature — the teacher labels these 'Bridges to Sub-Strand 1.4' and physically moves the cards to a new column titled 'OUR NEXT INVESTIGATION.' Students write in their notebooks: one sentence summarising the driving question answer, and one sentence framing their biggest remaining question.	periodic table help us predict what it will do?' — ask the whole class: 'Who wants to answer this as a full sentence — the best answer wins the unit?' Accept 3 volunteers. After the third, say: 'The periodic table is a map of electron arrangements — and electron arrangement is the hidden rule that governs everything we have studied.' Write this verbatim on the board. Then point to the open 'bridge' cards: 'These questions — about WHY Na^+ and Cl^- attract, and what holds NaCl together as a solid — are EXACTLY what Sub-Strand 1.4 will answer. You already have the right questions. That is the mark of a scientist.'		
Model Building Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Writing the Periodic Table (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student independently completes a structured summative task on a printed or handwritten answer sheet. The task presents an unknown element X with the electron arrangement [2, 8, 7] and states: 'Element X is found in the water treatment chemicals used at the Nairobi Gigiri water plant.' Students must: (1) State the group and period of element X in the periodic table; (2) State the number of valence electrons and predict X's chemical family (halogens); (3) Determine the type of ion X forms (X^-, anion) and its oxidation number (−1) and valency (1); (4) Write the electron arrangement of ion X^- in both shell notation and s/p notation; (5) Write the formula of the compound formed between X and potassium (K, electron arrangement [2,8,8,1]); (6) Write a balanced equation for K reacting with X_2 gas, including state symbols; (7) Extension: Explain in two sentences why element X and	Distribute task sheets face-down. Say: 'This is your individual opportunity to show what YOU understand — not your group, not your partner — you. Turn your sheet over when I say go. You have 20 minutes. Read every question before you begin.' Say 'Go.' Set a visible 20-minute countdown timer. At 10 minutes say: 'You should be on or past question 4.' At 18 minutes say: 'Two minutes — check your balanced equation has the same number of each atom on both sides.' At 20 minutes say: 'Pens down, sheets face-up.' Collect all sheets. Do NOT review answers aloud today — say: 'I will return these with feedback next lesson. What I can tell you is: if you could explain WHY the mystery element cards from Lesson 1 were confusing, and how you would sort them today — you have answered the driving question of this entire unit.'	Transfer Task (Summative Model Application) — by presenting an unseen element described in a Kenyan context (Gigiri water treatment), the task assesses whether students can transfer the unit's reasoning chain to a novel case. This mirrors the original anchoring phenomenon challenge (Mystery Element Cards) and closes the unit storyline with genuine resolution.	Observable indicators across all 7 items: (1) Group VII and Period 3 correctly identified; (2) Ion correctly written as X^- with oxidation number −1; (3) s/p notation for X^- written as $1s^2 2s^2 2p^6 3s^2 3p^6$; (4) Formula KX correctly derived; (5) Balanced equation: $2K(s) + X_2(g) \rightarrow 2KX(s)$; (6) Extension response references valence electrons and noble gas stability. Teacher uses a 4-point rubric: 0 = absent/incorrect, 1 = partially correct, 2 = correct with reasoning. Scores inform differentiated support at the start of Sub-Strand 1.4.

	element Y (electron arrangement [2,8,8]) behave so differently, despite both being in Period 3. Students work in complete silence. Teacher circulates without speaking, noting common error patterns for post-task feedback.			

D. TEACHER REFLECTION

After this lesson, reflect on the following questions before beginning Sub-Strand 1.4:

1. STORYLINE COHERENCE: When students annotated their Lesson 1 predictions, were the 'before → after' contrasts substantive (referencing electron arrangement) or superficial (referencing only memorised rules)? If superficial, plan a bridging review of electron arrangement in the first lesson of Sub-Strand 1.4.

2. POSTER QUALITY: Which of the six poster sections showed the most errors across groups? Common weak points are the labelled arrows between sections (students draw boxes but do not explain causal links) and s/p notation in the ion formation section. Use photographed posters to create a targeted worked example for the start of Sub-Strand 1.4.

3. GALLERY WALK FEEDBACK: Were students' 'Grow' notes scientifically specific, or did they default to aesthetic comments ('make it neater')? If the latter, revisit the modelling of criterion-referenced feedback at the start of the next sub-strand's first peer activity.

4. DQB CLOSURE: Were students able to articulate the driving question answer in a full sentence using 'electron arrangement' and 'periodic table position'? If fewer than 70% of the class could do so verbally, build a brief whole-class consensus sentence at the start of Sub-Strand 1.4 Lesson 1.

5. SUMMATIVE TASK PATTERNS: Review the 7-item task scripts before the next lesson. Key diagnostic indicators: (a) If >30% of students wrote X^{2-} instead of X^-, re-teach valency derivation from electron arrangement. (b) If >30% wrote an unbalanced equation for $K + X_2$, re-teach equation balancing with a particulate model. (c) If the extension question was widely left blank, the concept that noble gas configuration determines chemical inertness needs reinforcement in Sub-Strand 1.4.

6. EQUITY CHECK: Did all students — including those who were quieter in group work — have a genuine individual voice in the summative task? Cross-check task performance against participation records from Lessons 1–9. Identify any student whose task performance is significantly lower than their group contributions, and plan a one-on-one check-in.

7. BRIDGE TO 1.4: The open DQB cards (WHY do Na^+ and Cl^- attract? WHY is NaCl a solid?) are gold — display them prominently at the start of Sub-Strand 1.4 Lesson 1 so students see their own questions driving the next unit of study.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (in Lesson 1) that element cards for sodium and chlorine looked completely different — different reactivity ratings, different Kenyan contexts (ugali salt vs. Gigiri water treatment), different physical appearances — yet by the end of the unit they discovered these two elements combine to form the single most familiar compound in Kenyan daily life: the salt from the

	Malindi salt flats that seasons ugali across the country.
What did I learn?	Students learned that every element's behaviour is determined by its electron arrangement, which is encoded in its position on the periodic table (period = number of shells; group = number of valence electrons). This arrangement determines whether an atom forms a cation (loses electrons, e.g. $Na \rightarrow Na^+$) or an anion (gains electrons, e.g. $Cl \rightarrow Cl^-$), its valency and oxidation number, the formula of its compounds (using valency crossover), and the balanced equation for its reactions. The periodic table is therefore a predictive map, not merely a list.
How does this explain the phenomenon?	The mystery of the element cards is resolved: elements that seem physically different can be chemically complementary because their electron arrangements are mirror-image incomplete — sodium has one electron to give, chlorine has one space to receive. The 'invisible rule' (electron arrangement seeking noble-gas stability) explains every visible pattern on the cards and on the periodic table. This same principle — that electrons determine bonding — will drive Sub-Strand 1.4, where students will investigate WHY Na^+ and Cl^- attract each other and what that attraction means for the physical properties of substances.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.